GDP Across Time and Place

Are things getting better? How do we grow?

PROFESSOR KIRSTEN CORNELSON · GLOBAL MACROECONOMICS FOR MANAGERS

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Today’s plan

1 Yglesias article: are things getting better?

2 Poor place, rich place – a case study to drive your thinking

P1: In most economies, real output (GDP) follows a rising trend determined by the supply side of the economy

Three questions about Prop 1

1. What is the evidence in support of Proposition 1?

2. Why should we care, i.e., why does the trend growth rate matter?

3. What do we mean when we say the rising trend in GDP (i.e., growth) is "supply side determined"?

Our road map in context

We'll start with low-income countries and look at growth opportunities and challenges going up the income ladder.

Table O.1 World Bank country classifications and selected global indicators, 2022

INCOME CLASSIFICATION	SHARE OF GLOBAL POPULATION (%)	SHARE OF GLOBAL GDP (%)	SHARE OF PEOPLE IN EXTREME POVERTY GLOBALLY (%)	SHARE OF GLOBAL CARBON DIOXIDE (CO_2) EMISSIONS (%)
Low-income	8.9	0.6	36.5	0.5
Lower-middle-income	40.3	8.3	55.4	15.7
Upper-middle-income	35.1	30.3	7.1	48.6
High-income	15.7	60.8	1.0	35.2

“The most important question in all of economics”

Why are some places so rich and some so poor?*

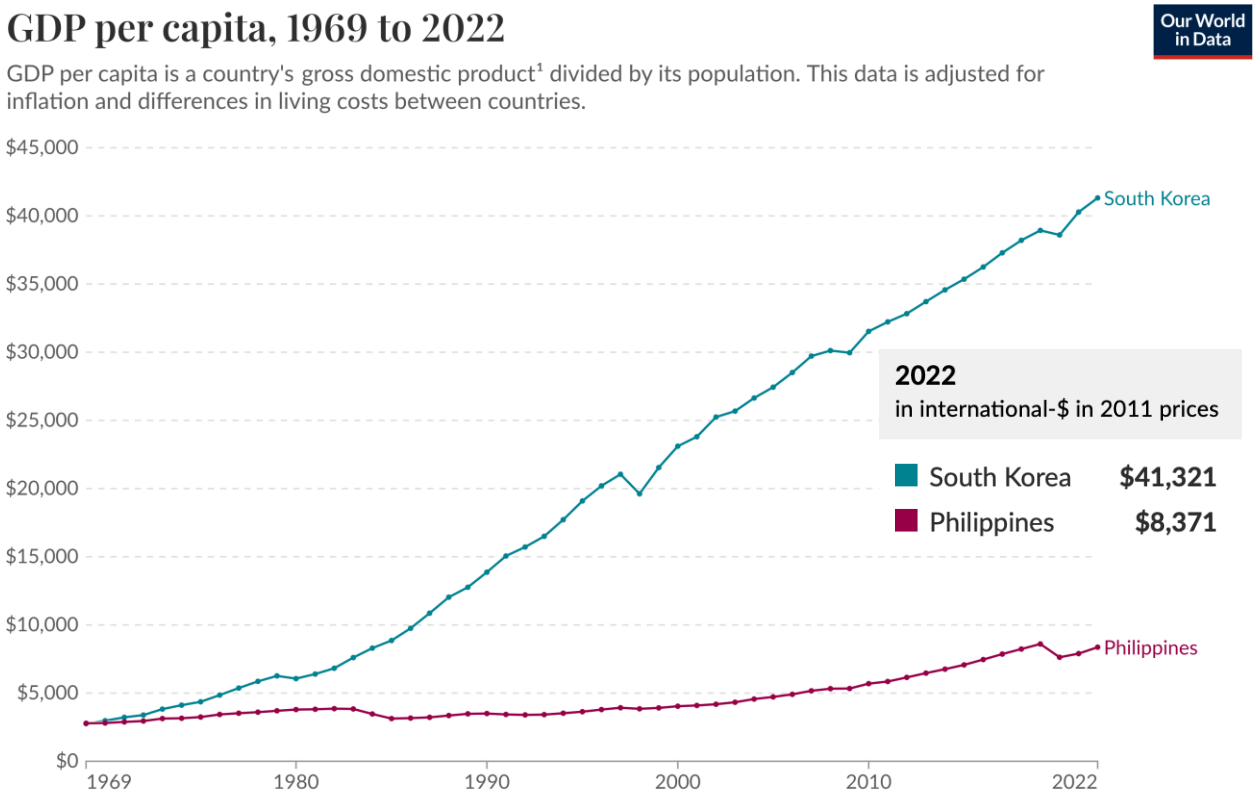
*Our specific focus: Why do some economies **grow faster** than others?

Choose a pay package

- Pay package A
  - Your employer places 1 penny on the first square of a chessboard. You get 2 pennies on the second square, 4 pennies on the third square, etc.
- Pay package B
  - Your employer places $1 million on the first square of a chessboard. You only use white squares: so you get $ 2 million on the next white square, $4 million on the third white square, etc.

Why the trend growth rate matters

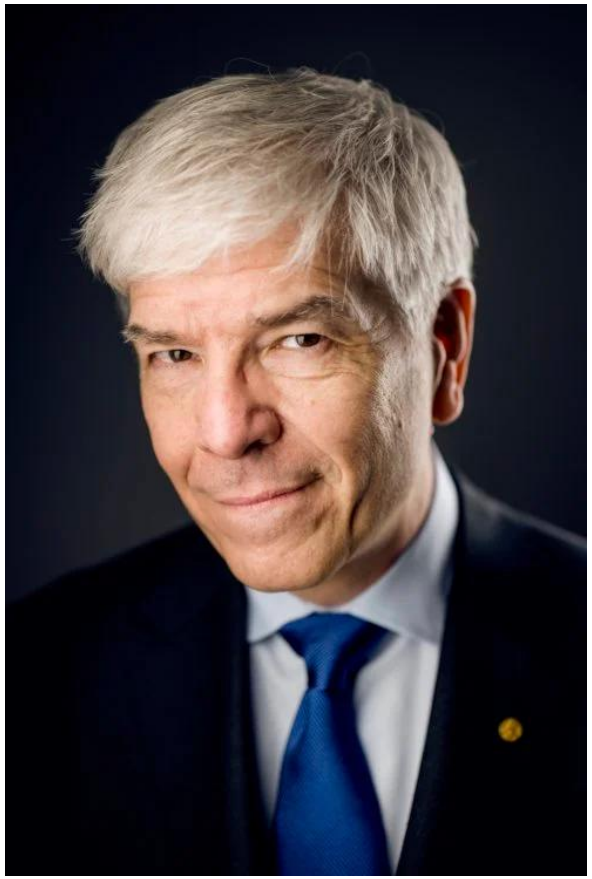
Human beings are reasonably good at estimating how things add up, but not when they compound. As a result, we often lose sight of how important even small changes in the average rate of growth can be.



A more relevant example

If 2 countries start out with the same level of real income and Country A grows 3% faster than Country B, in 55 years (about 2 generations), Country A will have a 5:1 ratio of superiority in income over Country B.

The growth guru speaks



Professor Paul Romer
Nobel Prize, 2018

"The choices that determine whether income doubles with every generation, or instead with every other generation, dwarf all other economic policy concerns."

A case study to drive our thinking

A tale of two countries: Canada and Vietnam

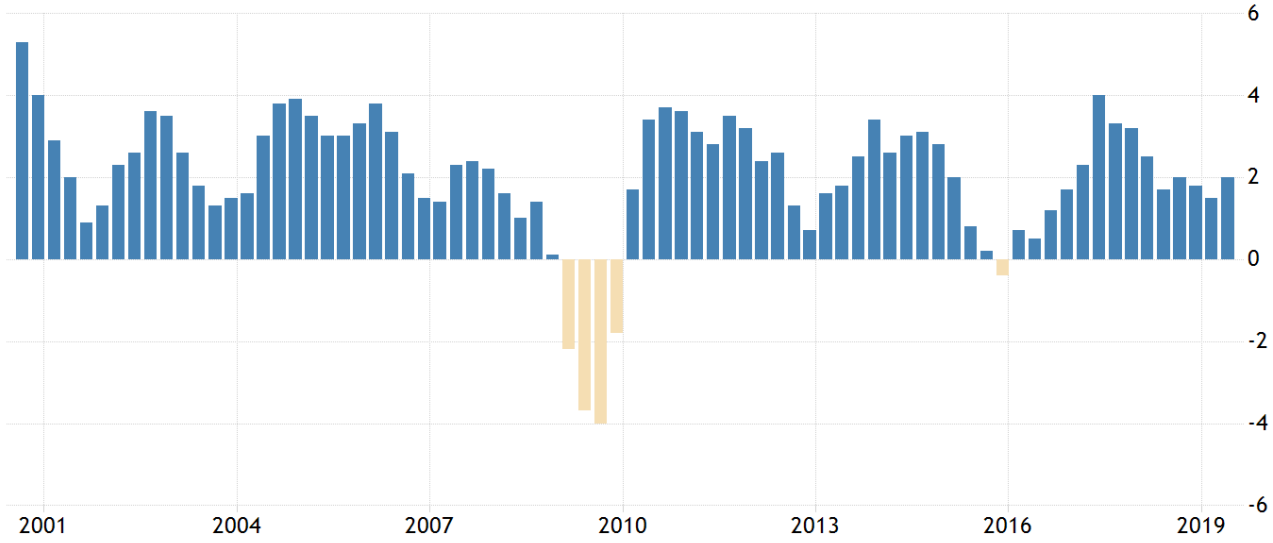




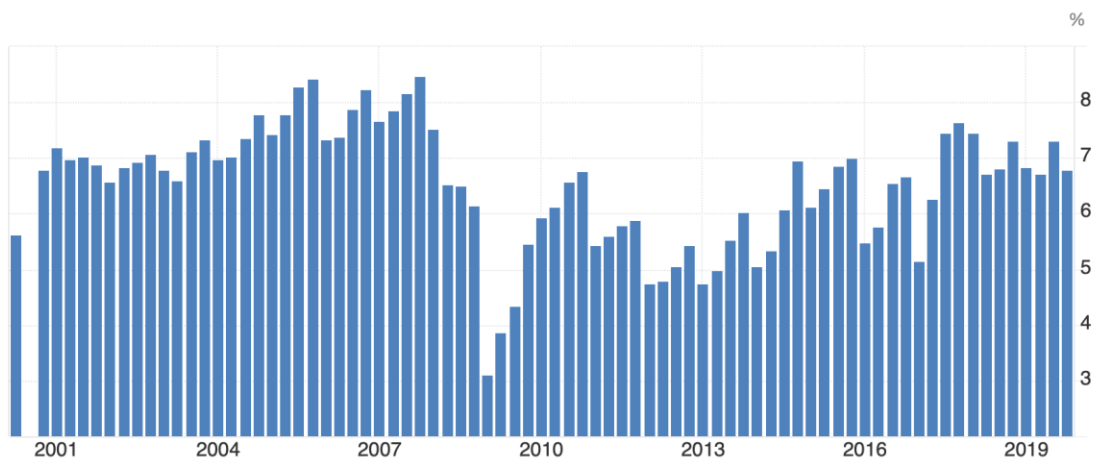
A tale of two countries: Annual GDP growth, 2000-2019

Growth rates over approximately one generation

Country A



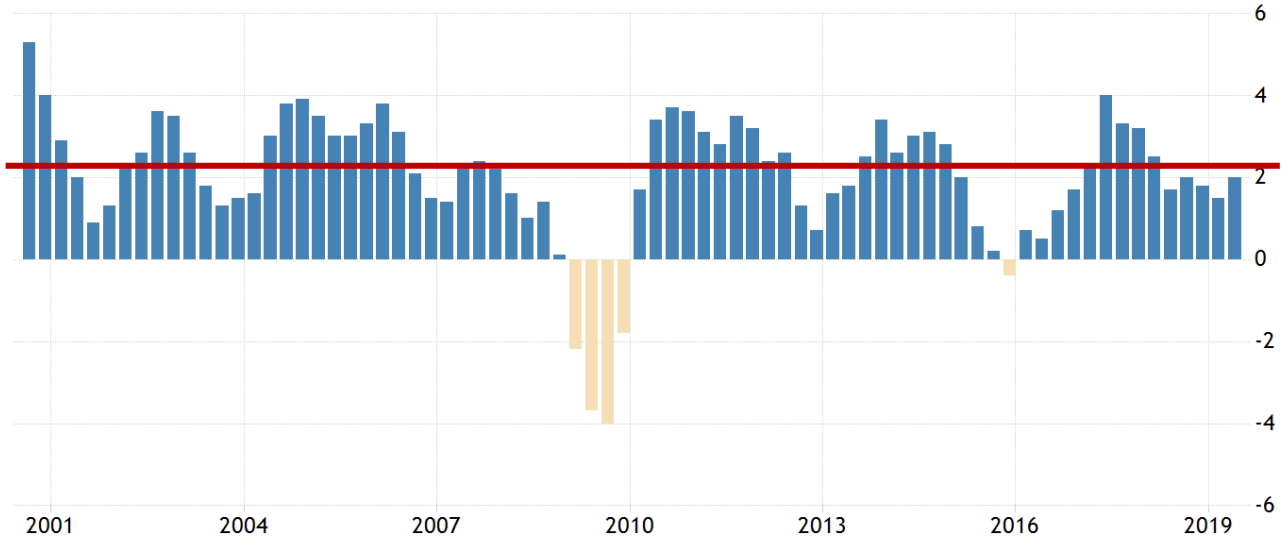
Country B



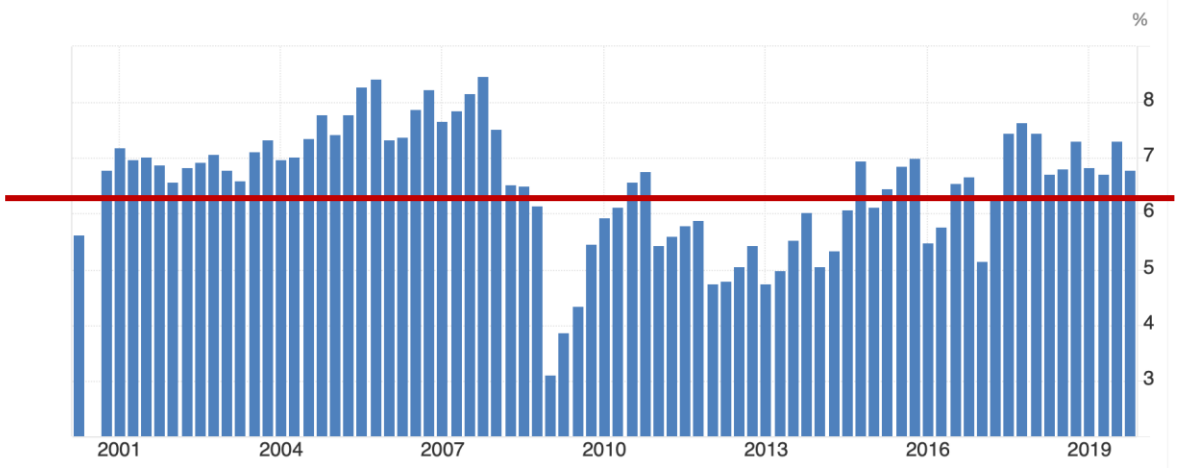
A tale of two countries: Annual GDP growth, 2000-2019

Is Canada country A or country B?

Country A



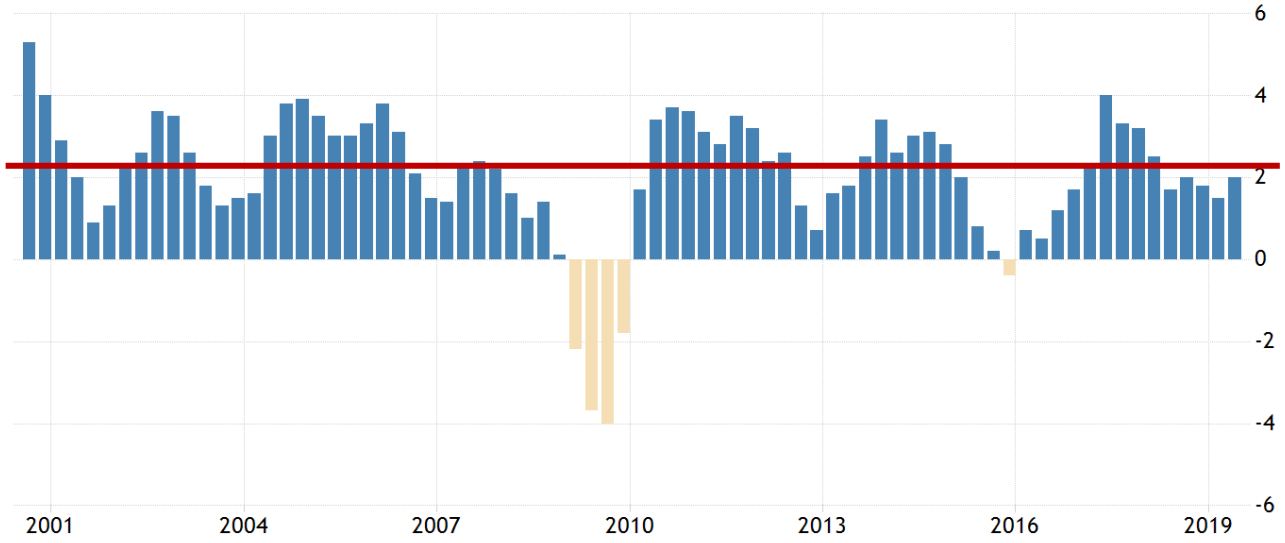
Country B



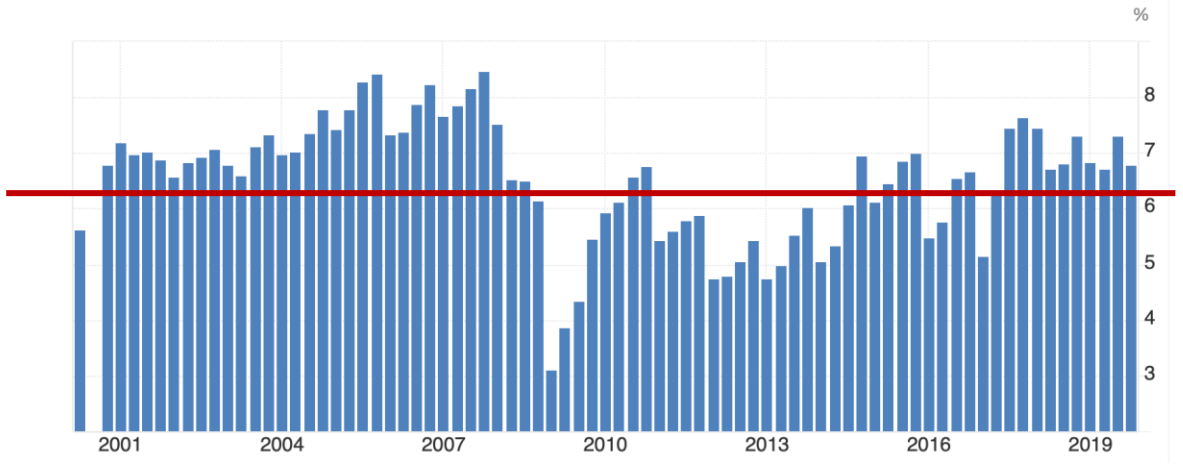
A tale of two countries: Annual GDP growth, 2000-2019

Is Canada country A or country B?

Canada: 2.3% CAGR

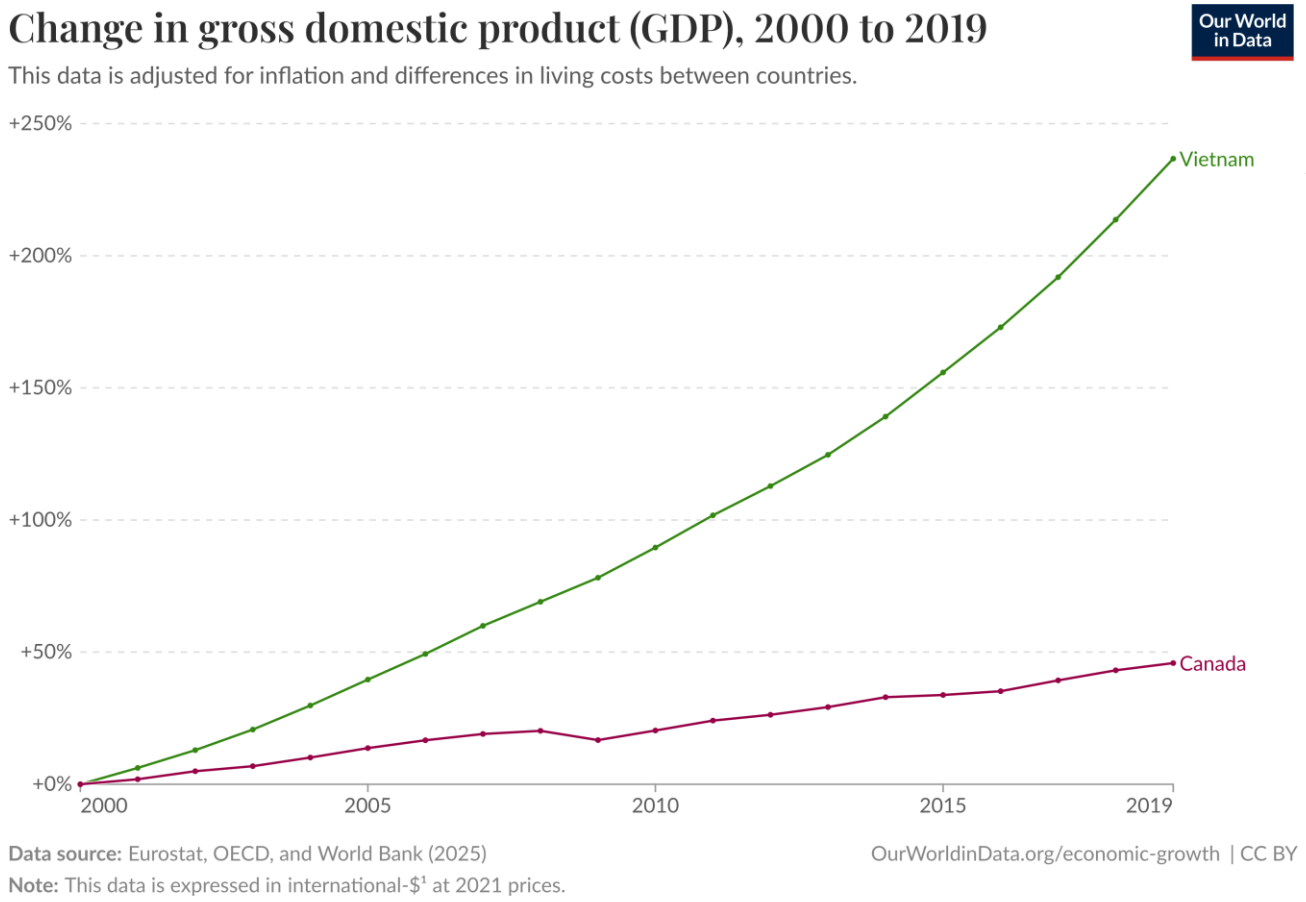


Vietnam: 6.3% CAGR



A tale of two countries: Canada and Vietnam

GDP growth over approximately one generation



236%

46%

Ivey

What this actually looks like on the ground

How has life changed for the average person in Vietnam?

Vietnam in 2000	Vietnam in 2019
Small village homes, limited electricity, clean water	Modern apartments, reliable electricity, running water
Dirt or gravel roads; motorcycles dominant, few cars	Highways, public transit, rising car ownership
Patchy access to healthcare and basic medicine	Expanding hospitals, improved health outcomes
Majority of labor in subsistence agriculture	Booming manufacturing sector, global exports
Very limited internet access, basic phones	Widespread smartphones, >70% internet penetration
Poverty rate >70%	Poverty rate <5%

Why has Vietnam had the higher trend growth rate?

- Develop a list of factors that support a Vietnam > Canada story on GDP growth over this 20 year period
  - I'm not asking you to do research on these countries!
  - I am asking you to think it through and develop some hypotheses

Close laptops.

Buzz groups for ~7 minutes

Explaining growth differentials: Canada v. Vietnam





Add capital, grow output

Early on in the process of economic development, a little capital goes a long way.







Behold, the future

“Any sufficiently advanced technology is indistinguishable from magic”

Arthur C. Clarke
Sci-Fi author

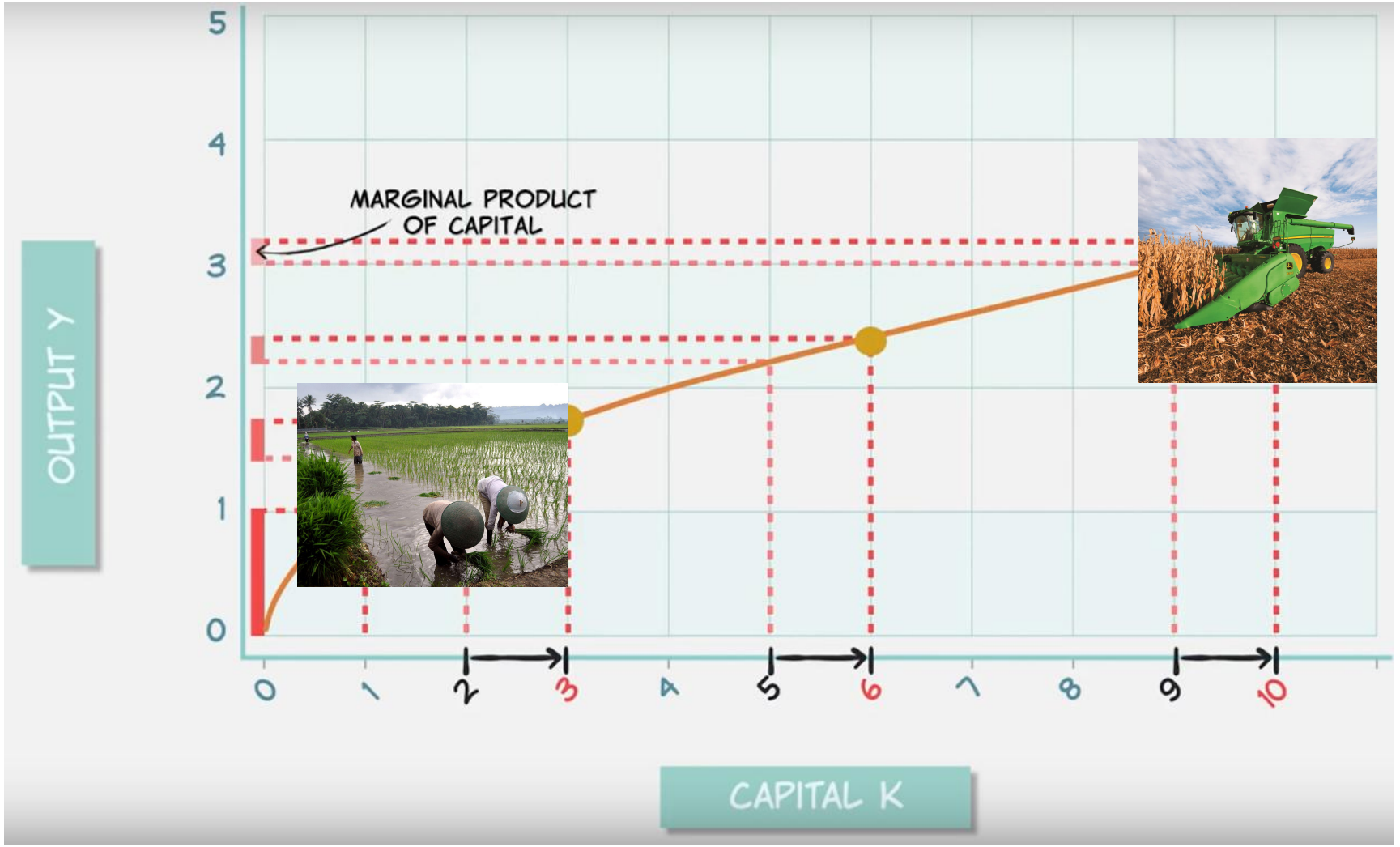


Get More in Rice

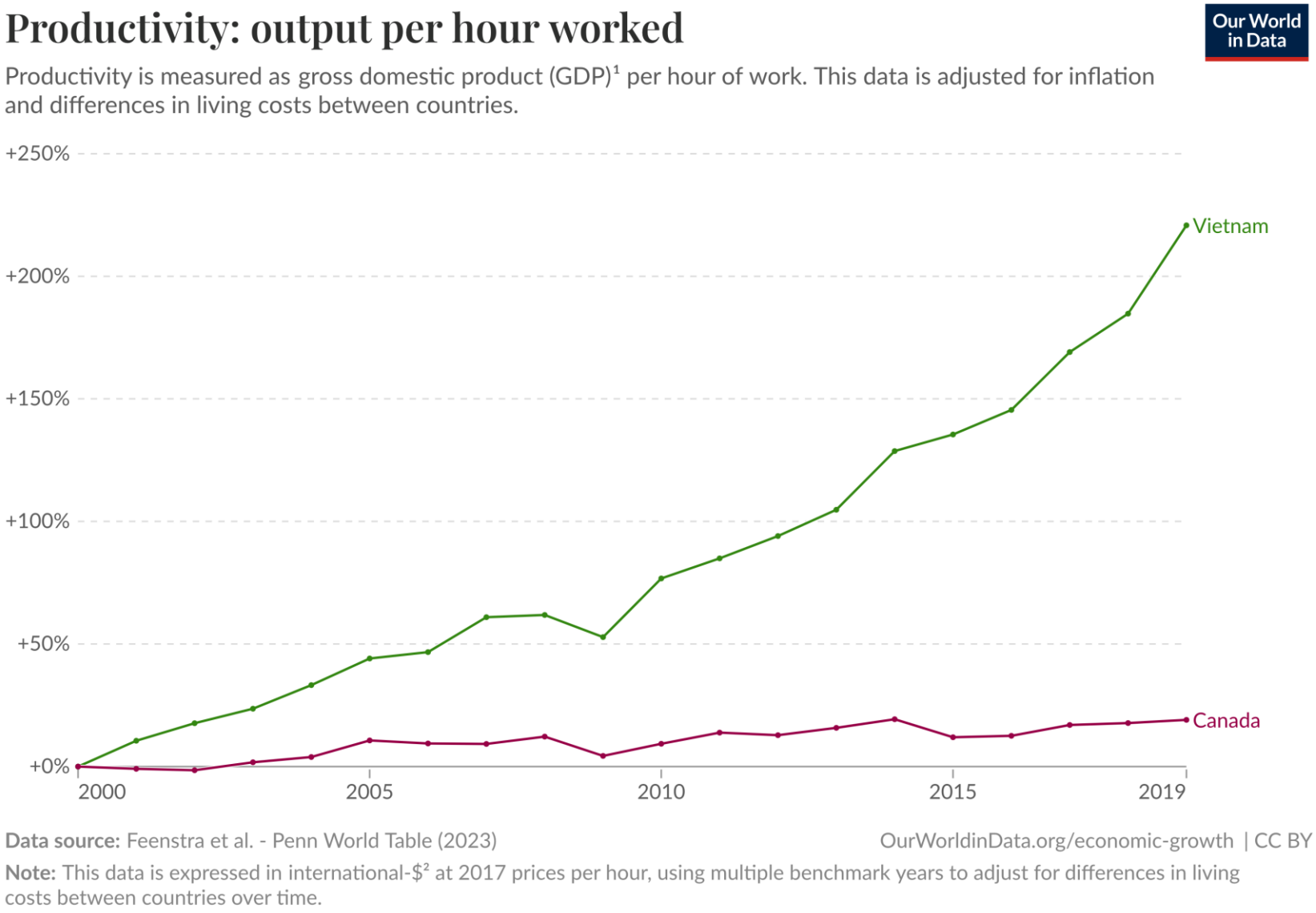
Gain resistance to the abrasiveness of rice, along with up to 65%[5] more capacity all while using 20% less fuel[6] with the X9 Rice combine package. Tougher coatings and more wear resistant components in key areas of the combine give enhanced wear resistance to tough hybrid rice, while still providing improved grain quality.

But… “iron law of diminishing returns”

For each added increment of capital, there is less and less output produced. The marginal product of capital declines. Growth becomes increasingly challenging through the pathway of “capital deepening”.



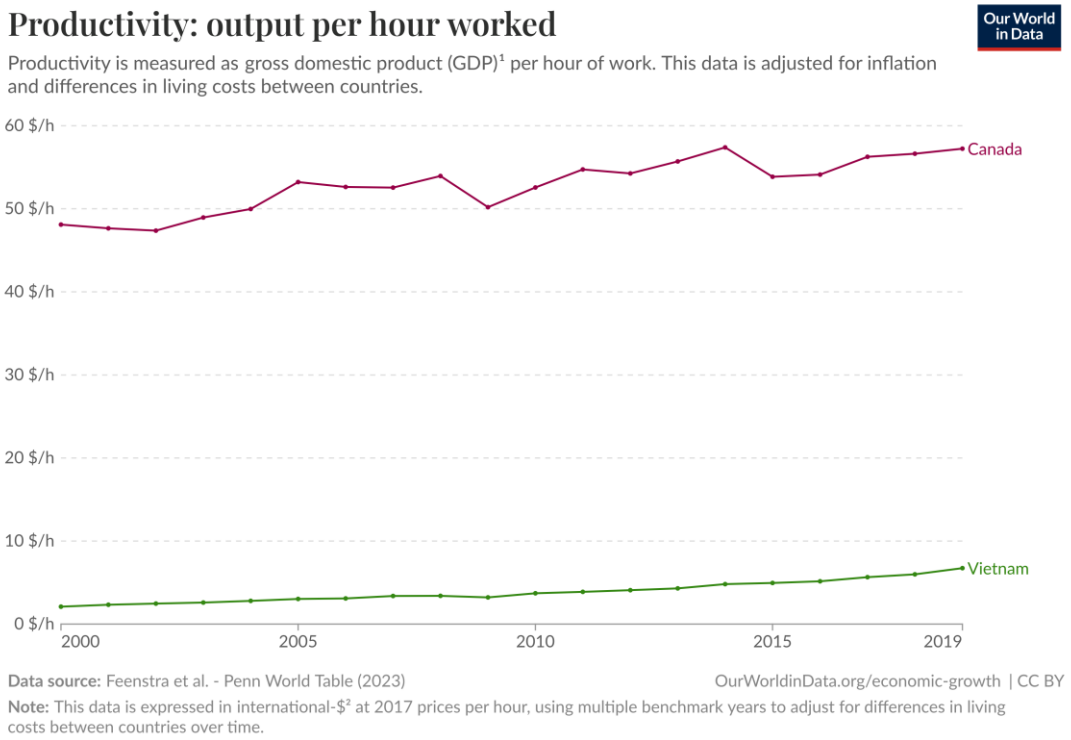
Add capital, grow ~~output~~ productivity



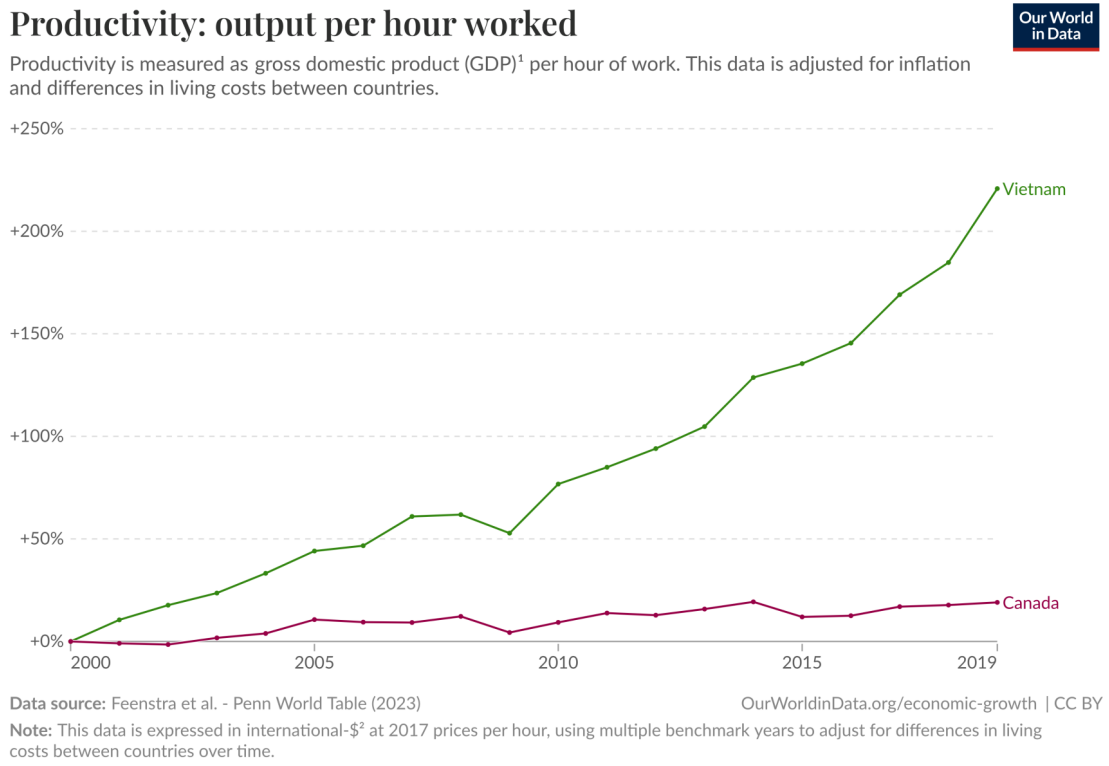
Don’t confuse *level* and *growth* of productivity

Canada is a much richer country because of our higher productivity (level). Vietnam is catching up (growth)

Level



Growth (△ Level)

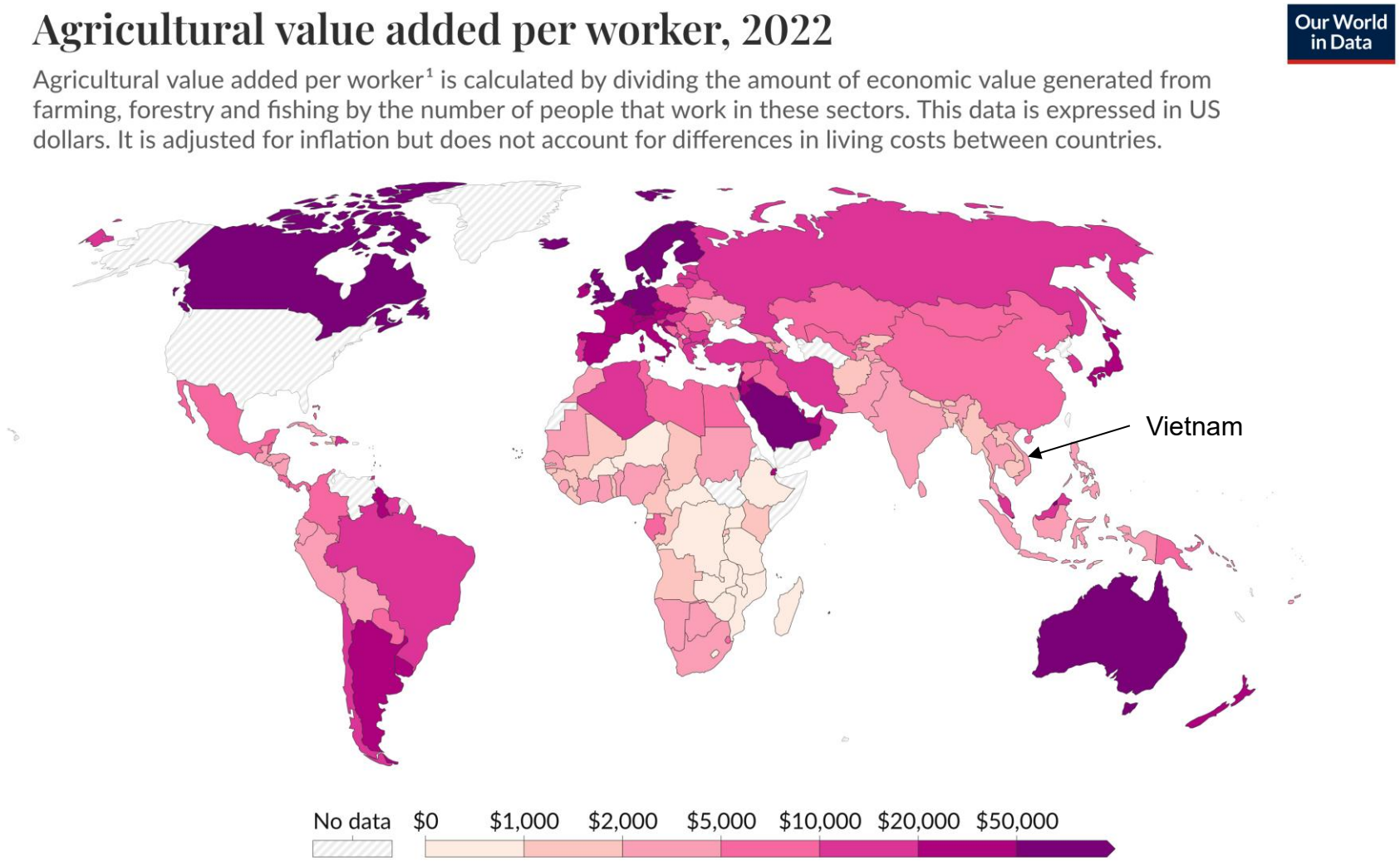


Canada in 2019:	$57.20 / hr
Vietnam in 2019:	$6.70 / hr

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Productivity in Agriculture

An average agricultural worker in Canada produces ~$125,000 per year of value. In Vietnam, ~$2,000



Agriculture in Canada and Vietnam

Spot the pattern. How does this fit into our emerging story of how economies develop and grow?

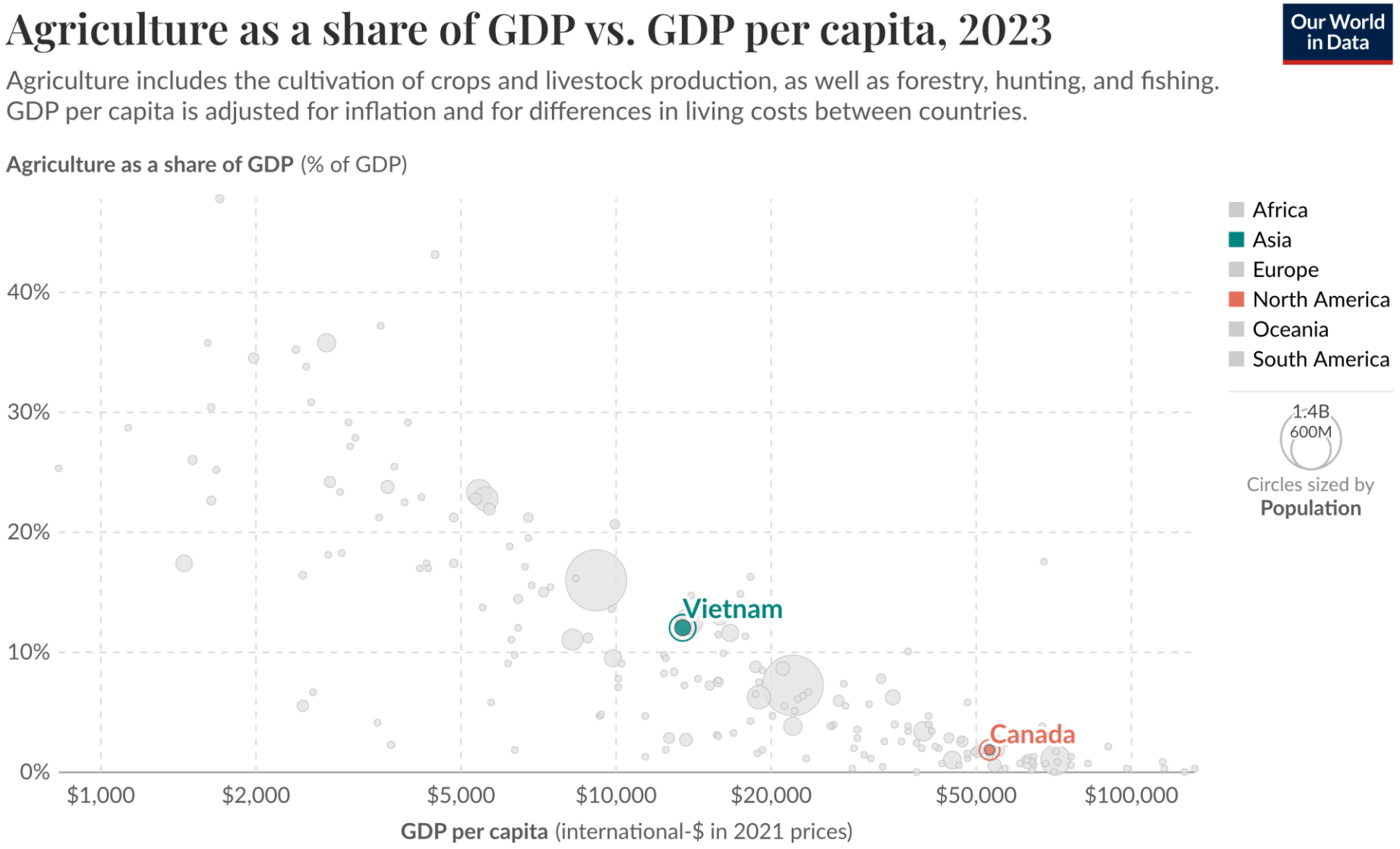
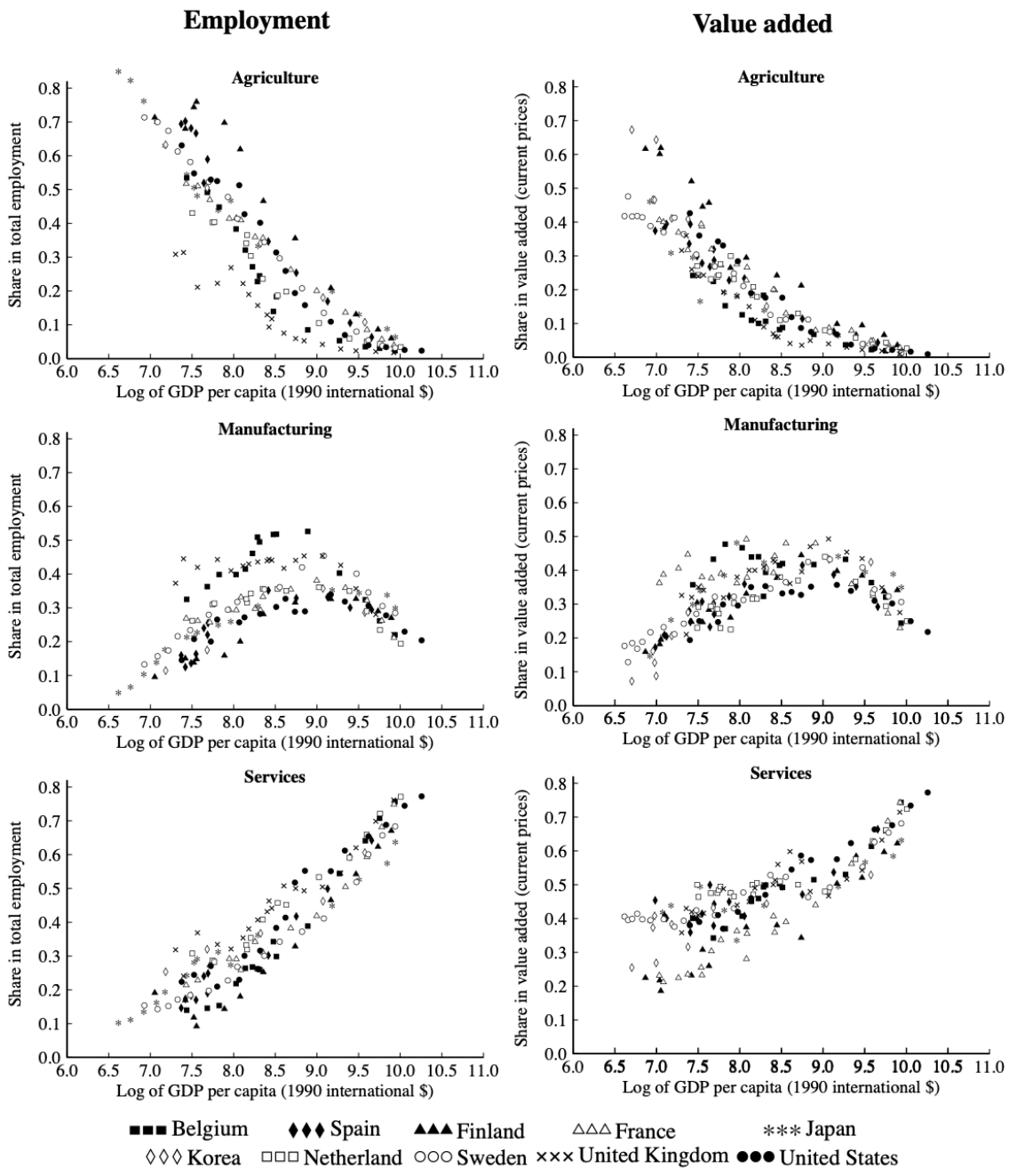


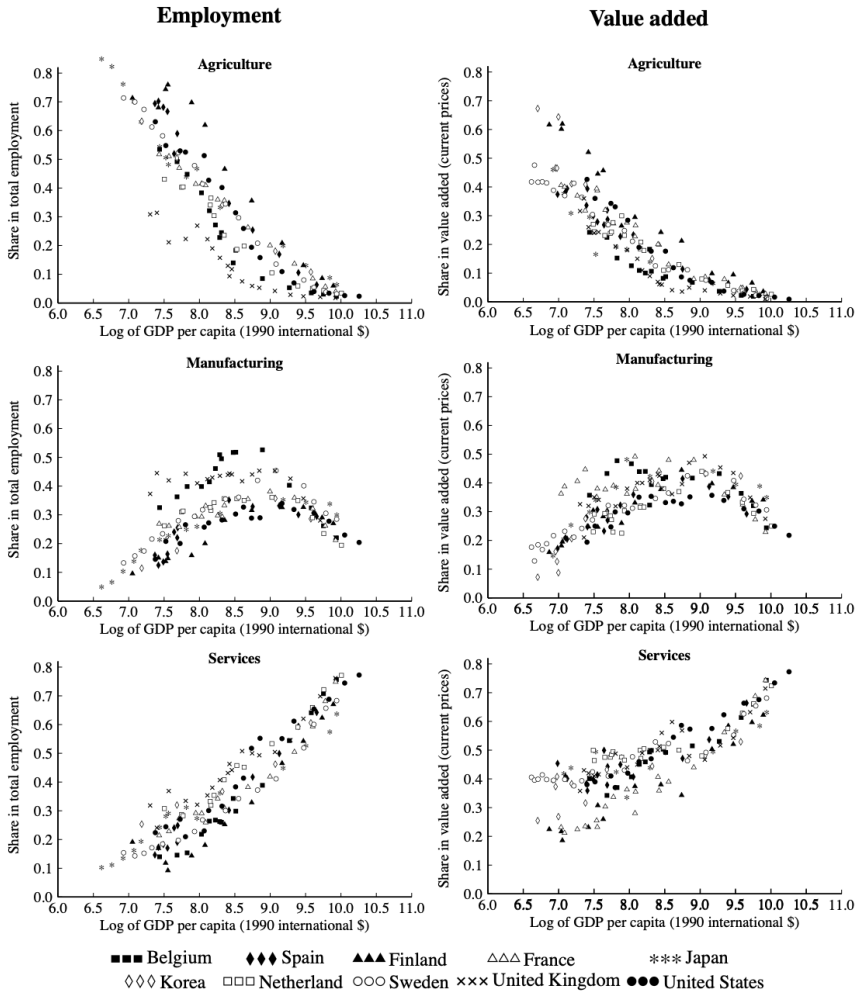
Figure 27.1: Structural Transformation: Figure 1 in Herrendorf et al. (2013)



Structural transformation: primary, secondary, tertiary

Workers and investment away from agriculture to manufacturing and services, which tend to have higher productivity because of greater capital, specialization and scale, and faster technological progress.

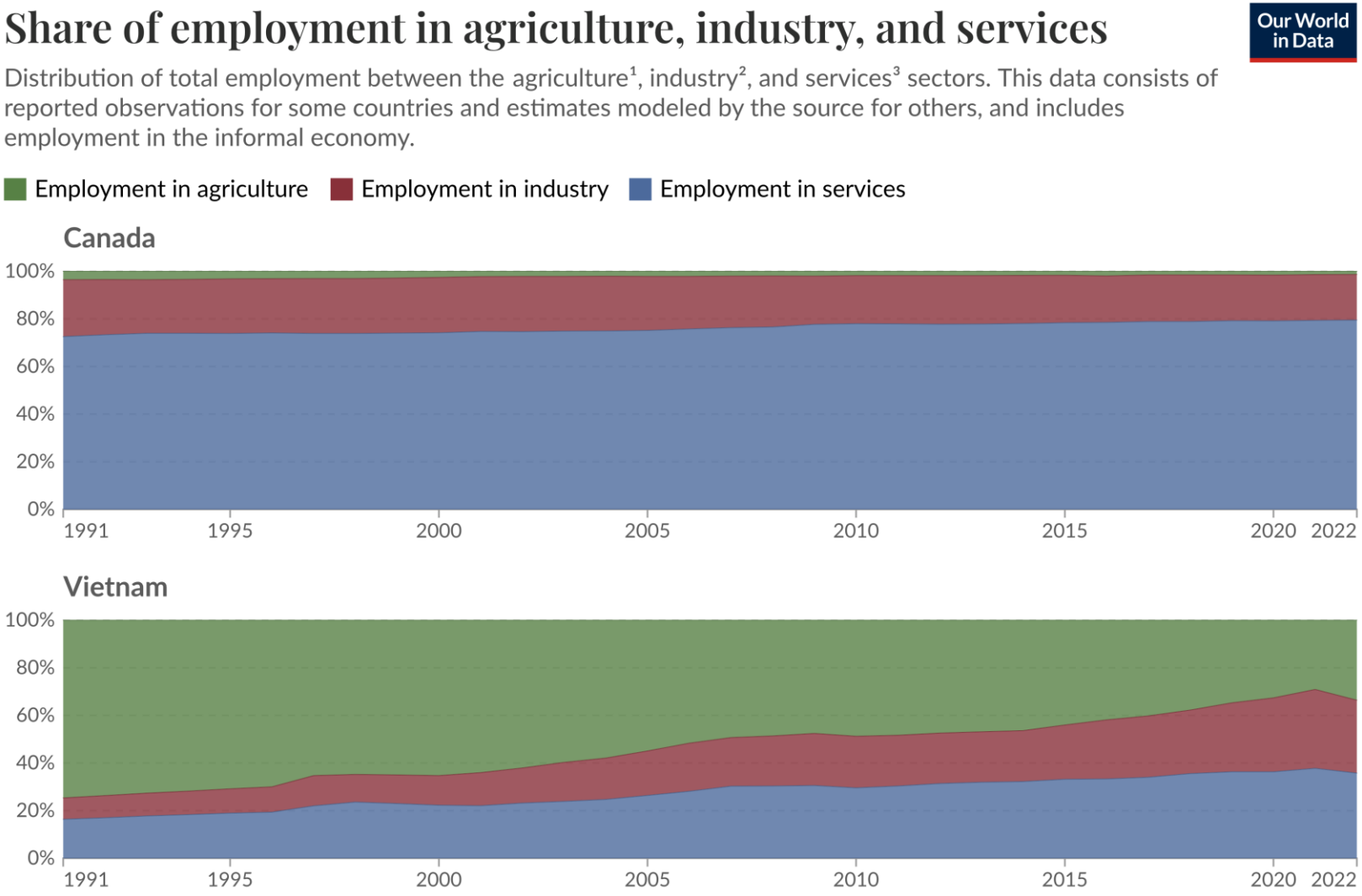
Figure 27.1: Structural Transformation: Figure 1 in Herrendorf et al. (2013)



- Note that this is (mostly) a capitalistic process: people and investment gravitate to sectors with higher incomes and better ROI. Government policy can, of course, facilitate the transformation.

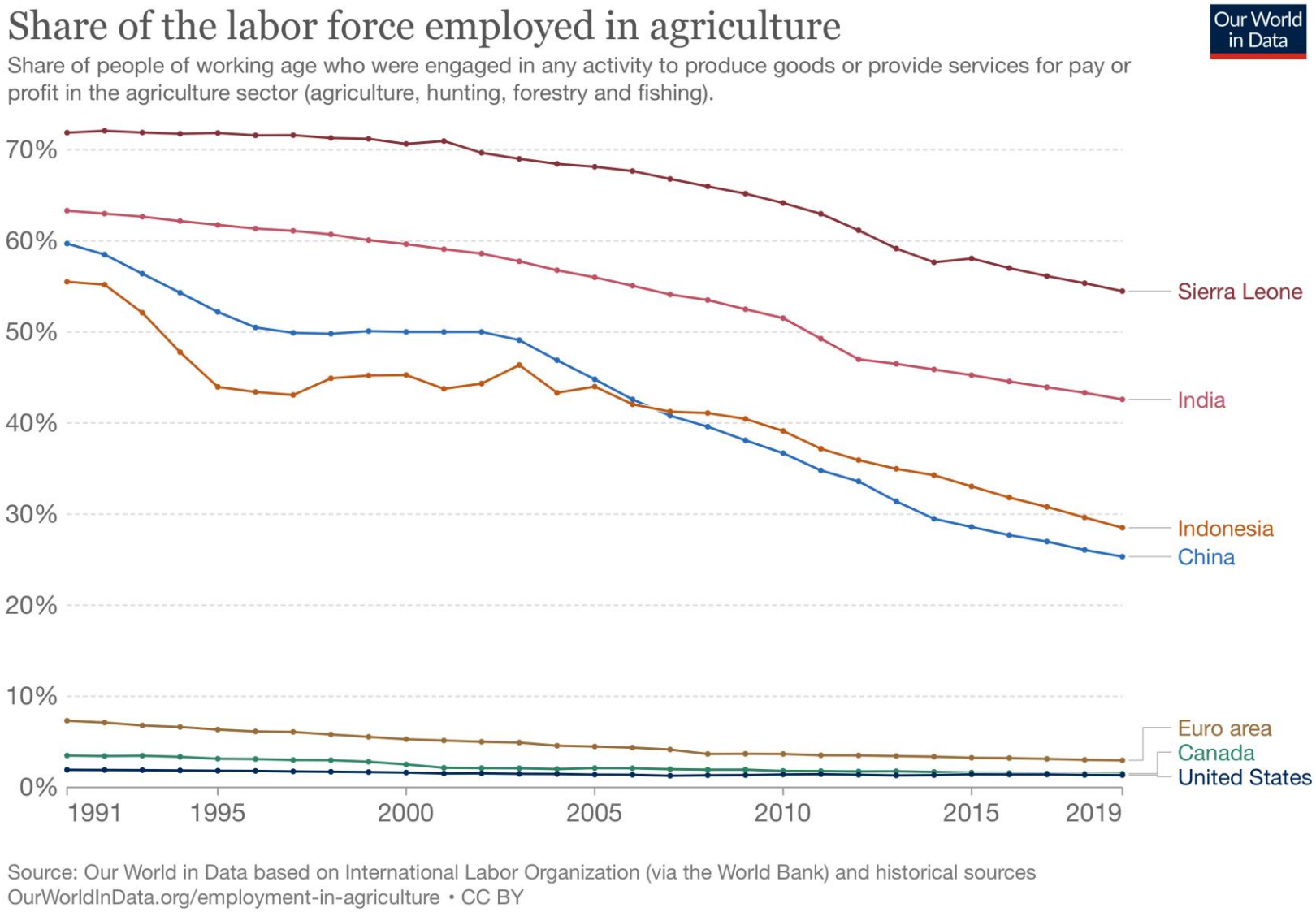
Structure of industry in Canada and Vietnam

Services are us. Nearly 80% of our economy today is services. In Vietnam, it is about 36%, up from 16% in 1990. Back then, agriculture employed nearly 80% of Vietnamese people.



Farmers are them

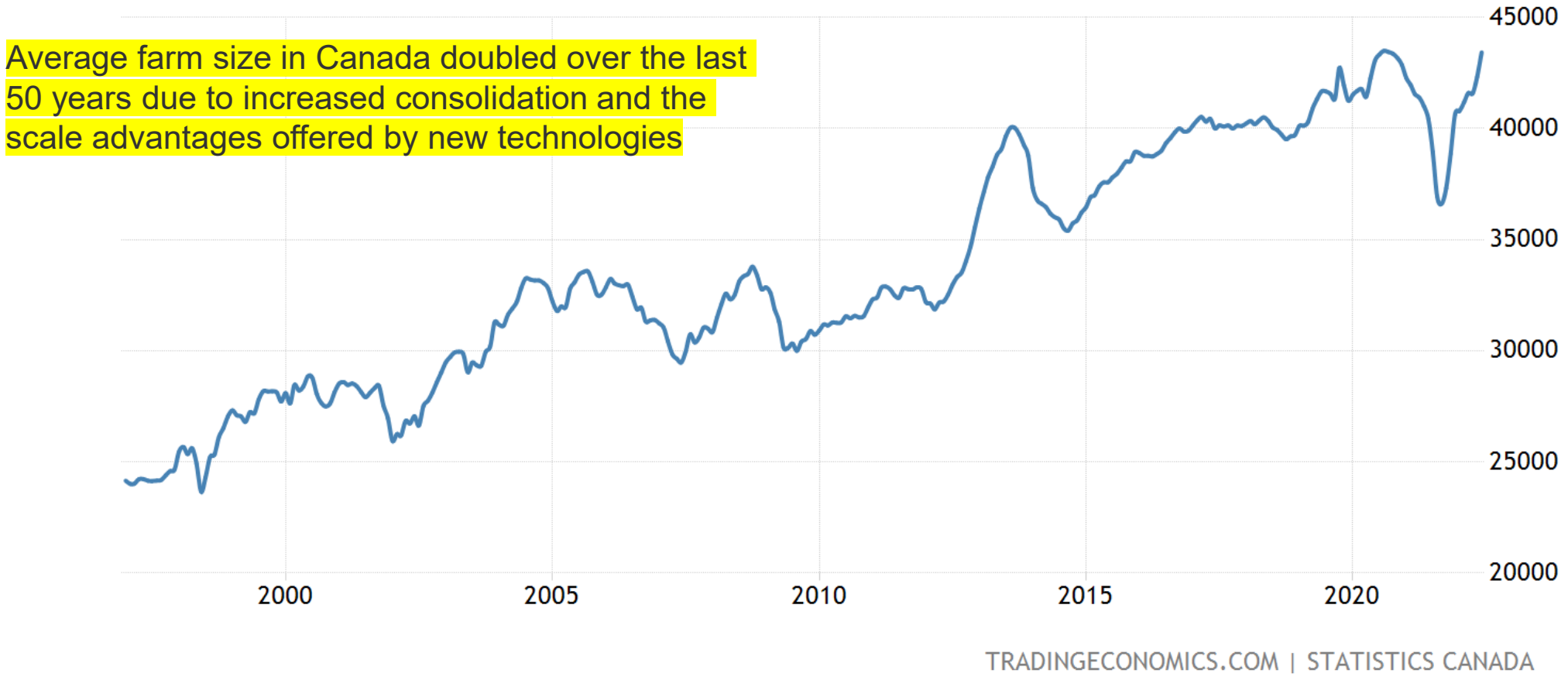
The majority of people in the poorest countries work in agriculture. In the rich countries, hardly anyone does.



Total agriculture output may rise indefinitely

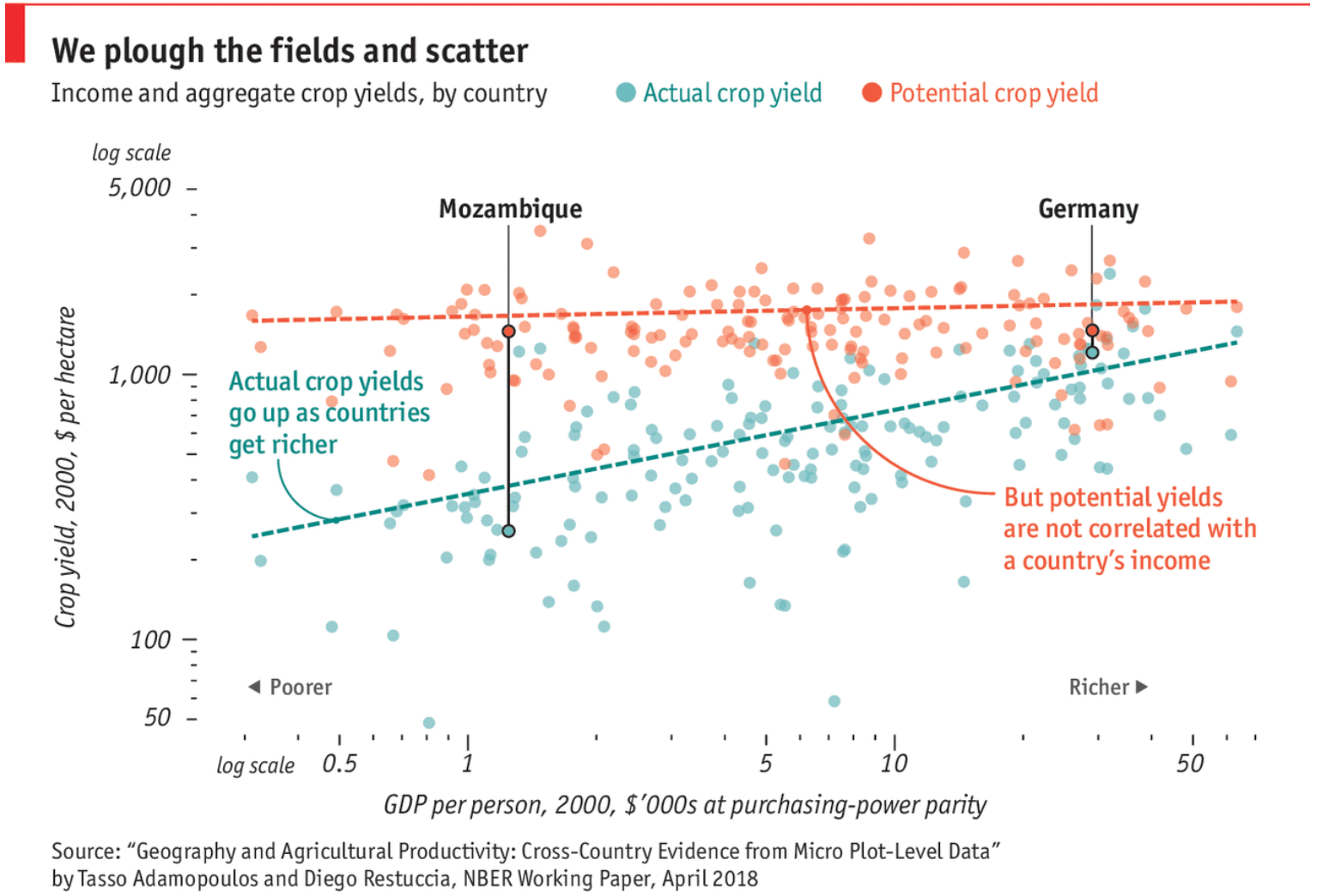
The number of farmers in Canada has declined to less than 250,000, but total output continues to rise. Productivity growth (increases in output per farmer) drives this pattern.

Average farm size in Canada doubled over the last 50 years due to increased consolidation and the scale advantages offered by new technologies



Closing the gap between actual and potential

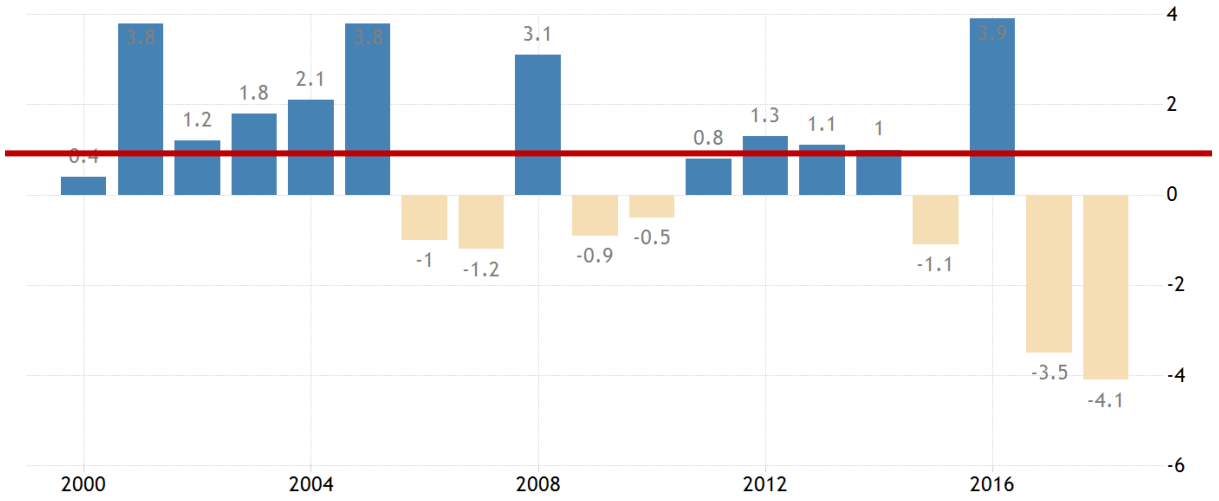
Agricultural productivity is not about land quality and climate. Geography matters, but not as much as you think. It is about capital, innovation, infrastructure, public policy.



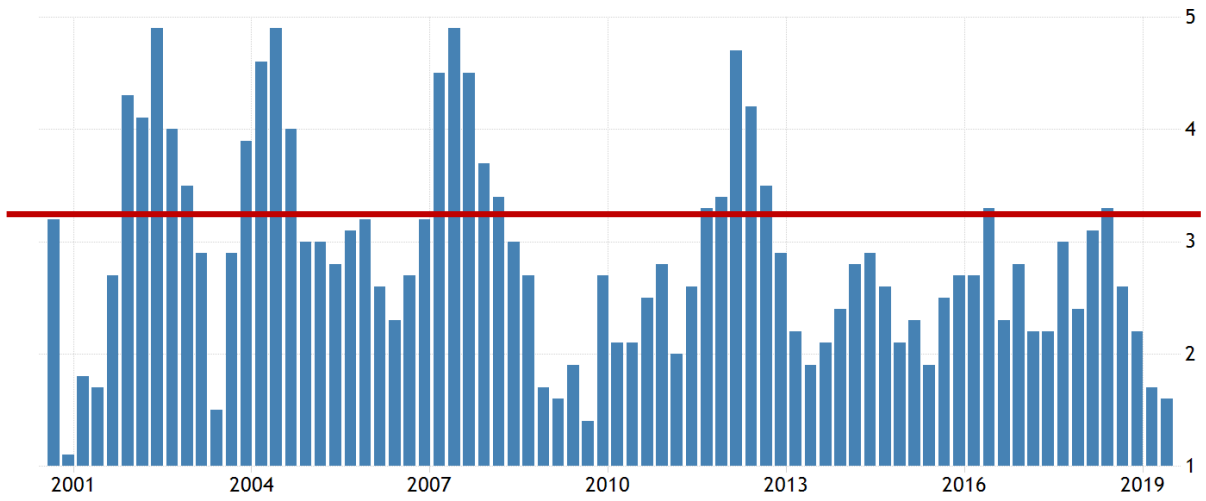
Source: "Geography and Agricultural Productivity: Cross-Country Evidence from Micro Plot-Level Data" by Tasso Adamopoulos and Diego Restuccia, NBER Working Paper, April 2018

A point to remember: being poor is no recipe for growth!

North Korea



Australia



North Korea at Night



Probably bad for business



Also bad for business…coups, civil wars and invasion

Note that “political instability” doesn’t capture the problem – you could call North Korea “stable”, but also deeply repressive. Growth and development happen when political institutions broadly are **effective and growth-enhancing;** both chaotic instability and rigid, extractive stability can block economic progress.

How leaders in Africa left office, 1960-2000

Decade	Through coup war or invasion	Elections
1960-69	27	0
1970-79	30	0
1980-89	22	1
1990-2000	22	14

Source: Arthur, 2000: “Risk, Rule and Reason: Leadership in Africa”, Africa Notes, Institute for African Development, Cornell University.

Why are some places so rich and some so poor?

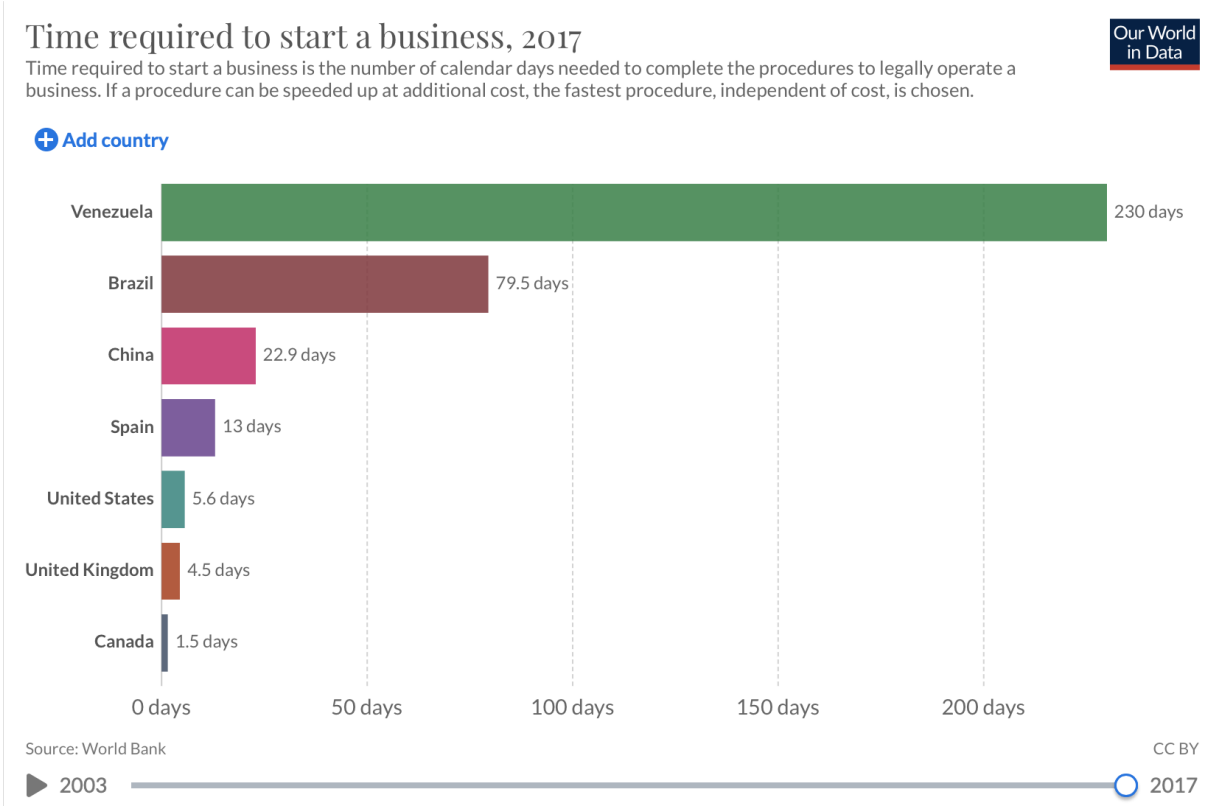
An important but incomplete answer – "institutions matter".

Institutions Matter

Property rights refers to the ability of people and businesses to own land and capital. Ownership enables people to produce, buy, and sell goods and services and to profit from business ventures. The right to own private property also includes the ability to sell that property. Without secure property rights, not many people would be willing to start a business, buy a house or land, or invest.

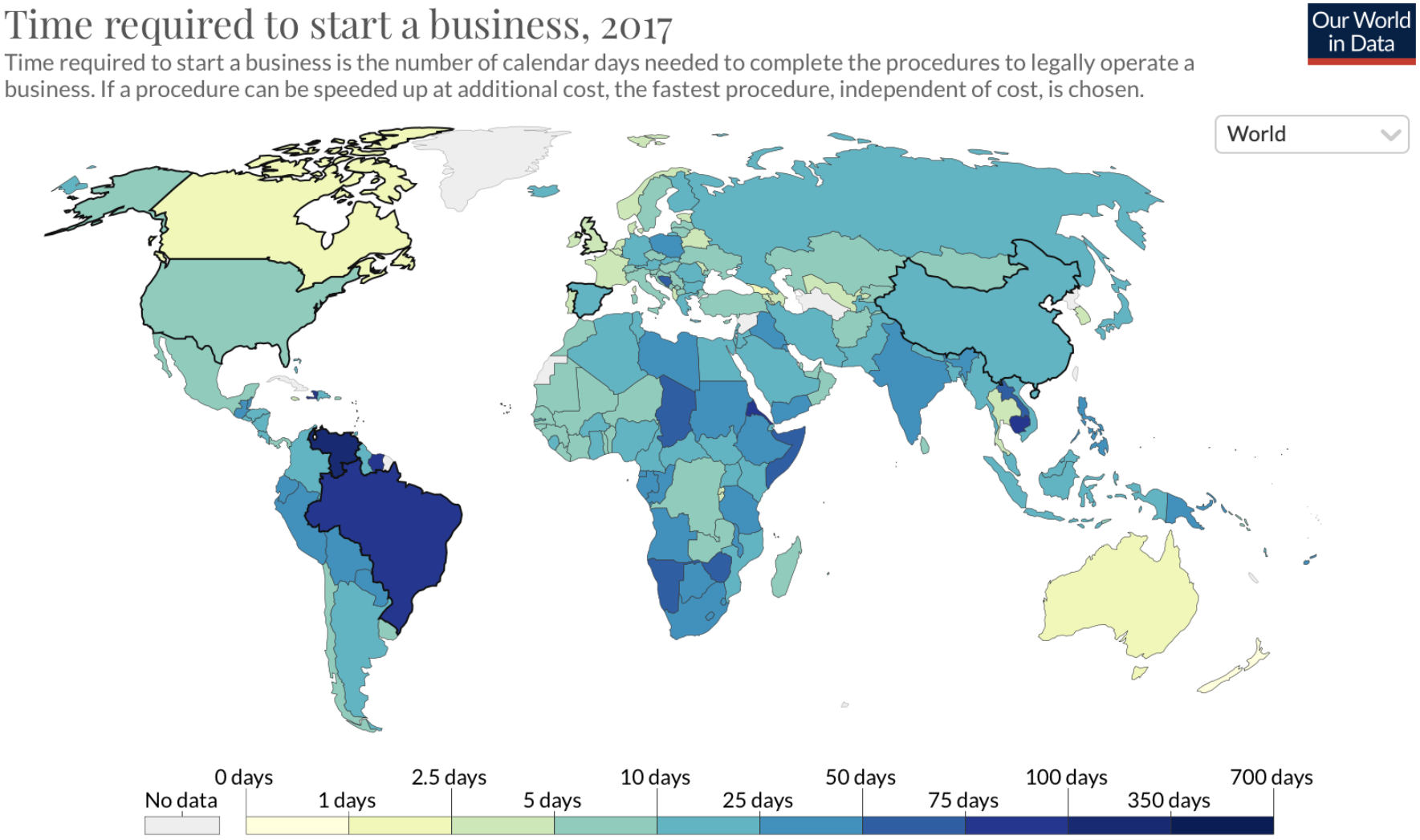
Free and open markets refers to the ability of people and businesses to buy and sell goods and services with minimal interference by government. This is a balancing act: While government provides protection for its citizens through its regulations, too much regulation can make economic transactions unprofitable and unattractive.

Rule of law holds that the law—not individual government leaders—governs a nation and that the government, government leaders, and all people must follow the law. The rule of law provides a sense of stability and certainty for economic transactions. For example, people and businesses are more likely to invest for the future if they feel confident that the "rules of the game" will be stable, rather than in a state of constant change.



Why are some places so rich and some so poor?

Clearly "institutions matter". But how to get them and maintain them…well, that is difficult



Vietnam took the “good institutions” pill in 1986

Vietnam’s takeoff story story begins with the set of institutional changes known as the Đổi Mới reforms. Today, the average person in Vietnam is **roughly 10x richer** materially than they were in 1986.

Pre-1986

- Centrally planned economy with state control over production
- No private sector or property rights
- Closed to international trade and investment

Post 1986

- Decentralization in agriculture: land-use rights granted to farmers
- Legal reforms supporting property rights and private enterprise
- Opening to trade and foreign direct investment (FDI)
- Infrastructure expansion, continued legal reforms
- Adaptive, incremental 'trial-and-error' policymaking

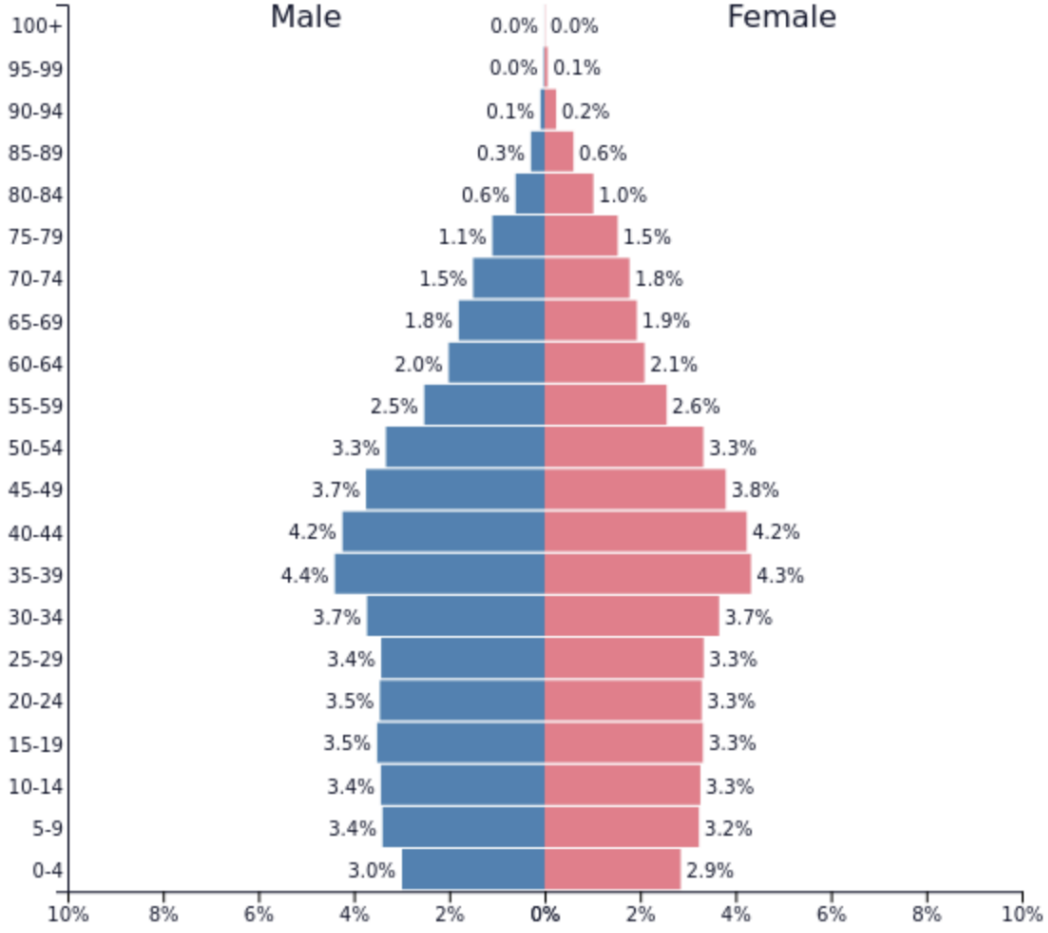
Hey, let’s talk about babies!

They’re pretty cute. Good for the economy, too.



Canada and Vietnam: Population Pyramid, 2000

How do babies fit into your story about the drivers of economic growth?



Canada – 2000
Population: 30,891,803

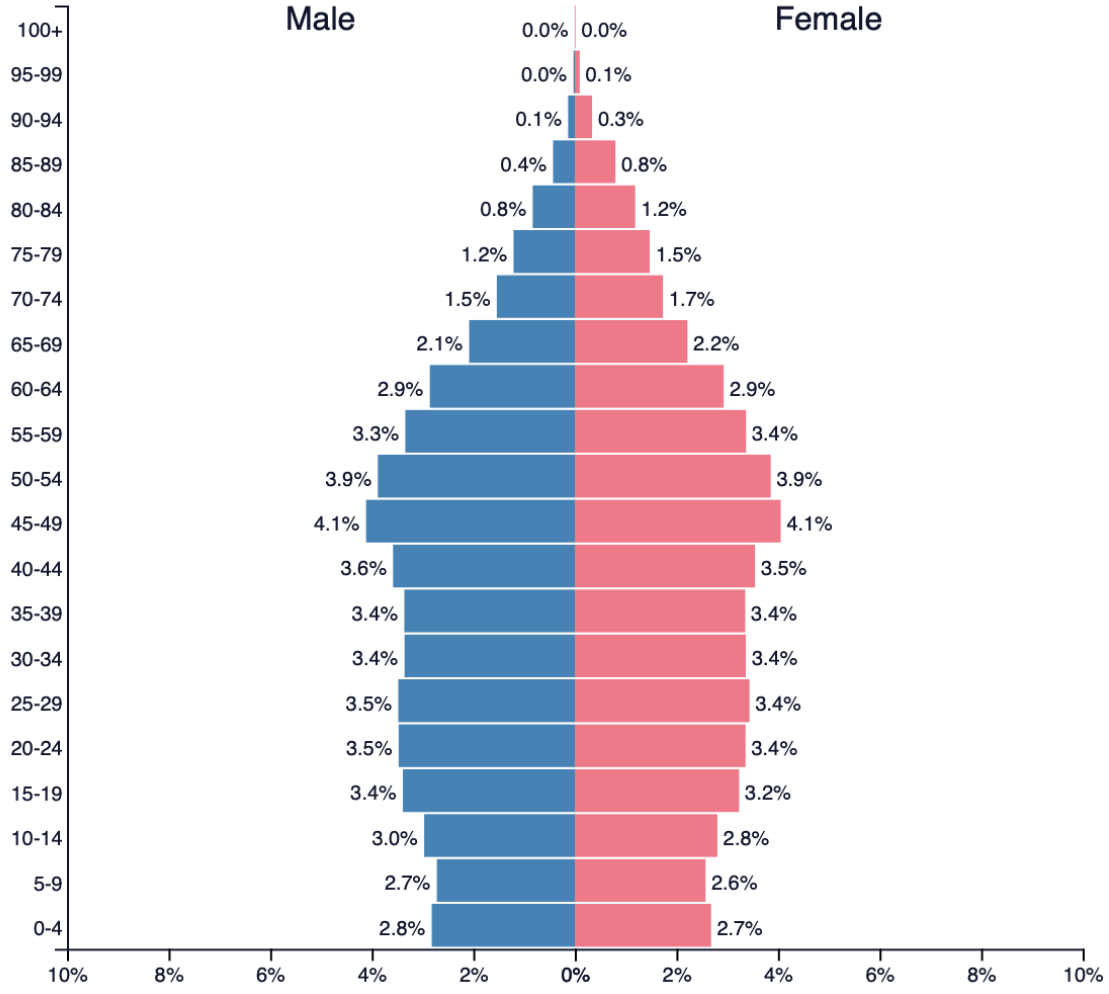


Vietnam – 2000
Population: 77,154,011

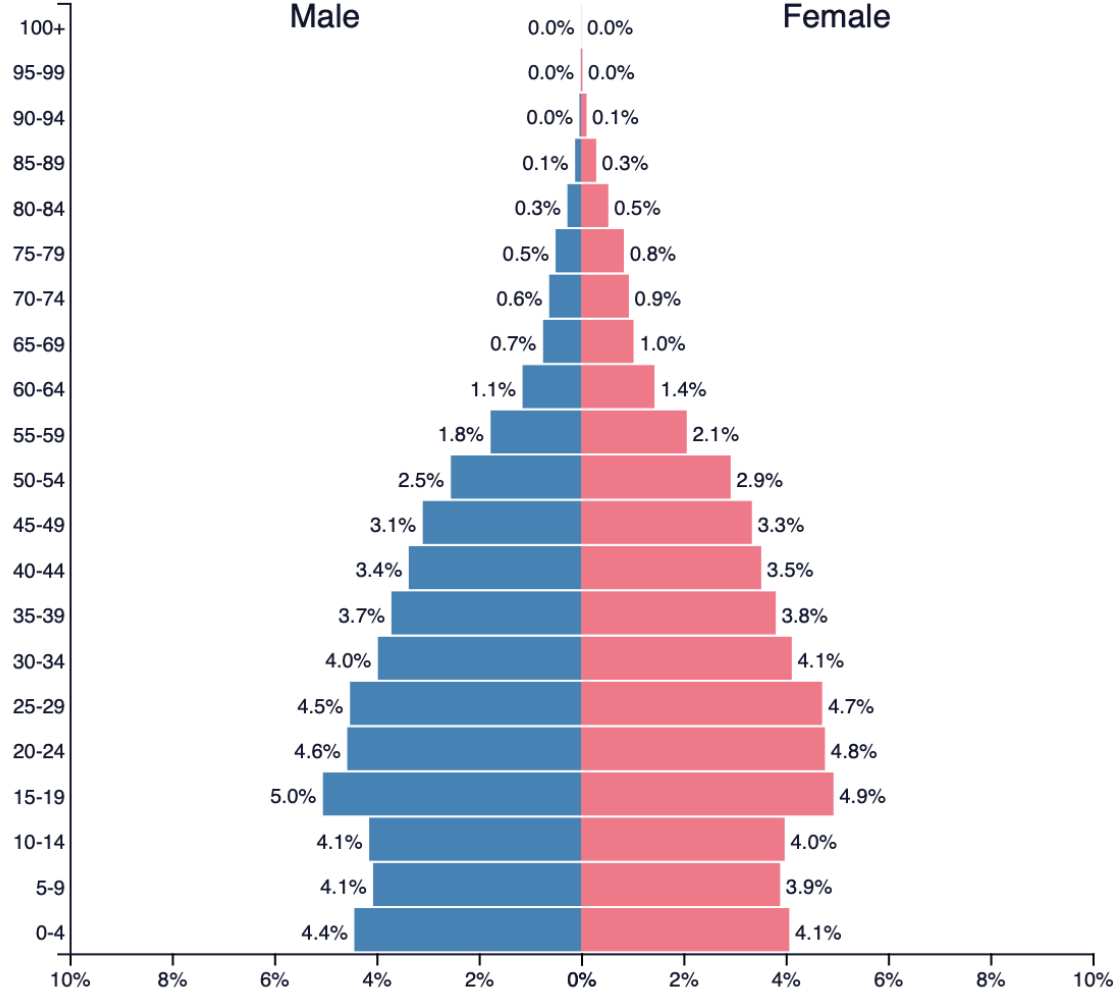
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Canada and Vietnam: Population Pyramid, 2010

The demographic sweet spot: lots of working age people, not too many “burdensome” oldies (lol, jk)



Canada – 2010
Population: 34,196,898

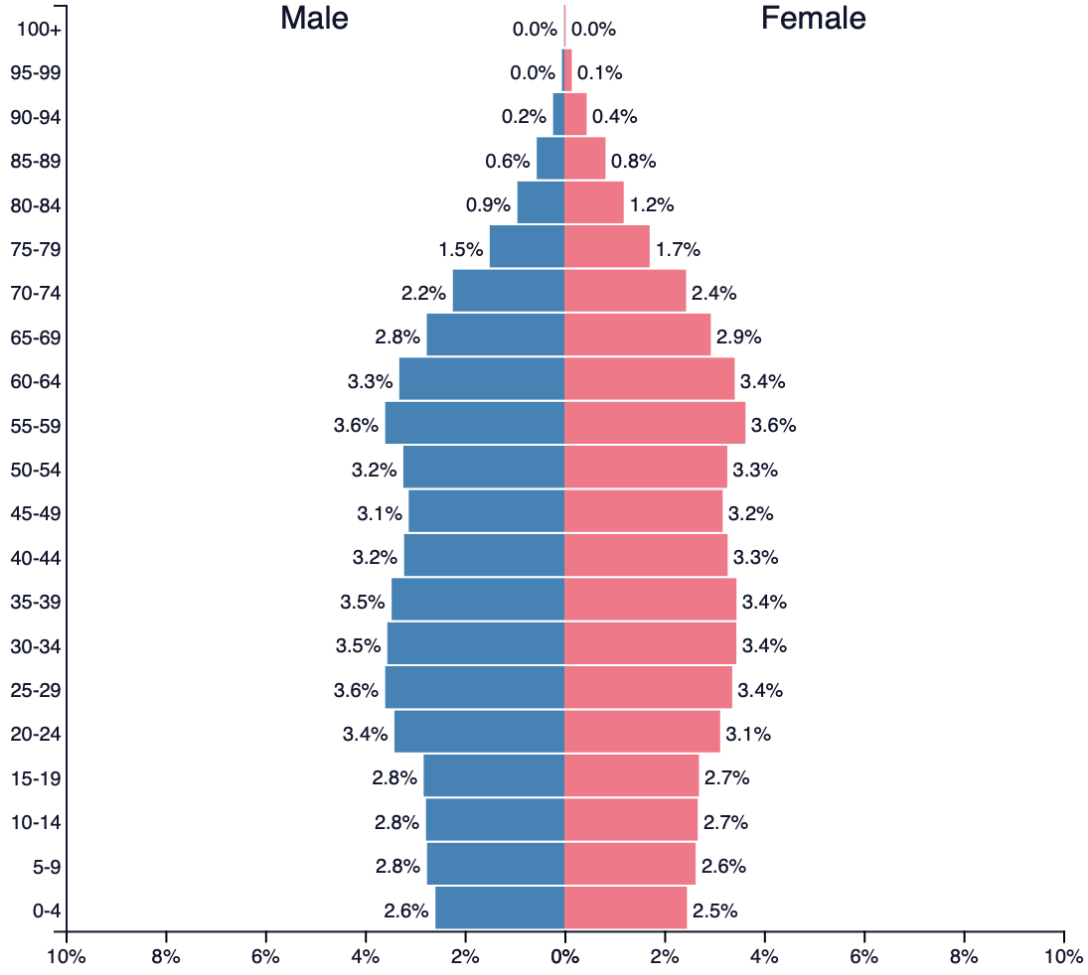


Vietnam – 2010
Population: 87,455,110

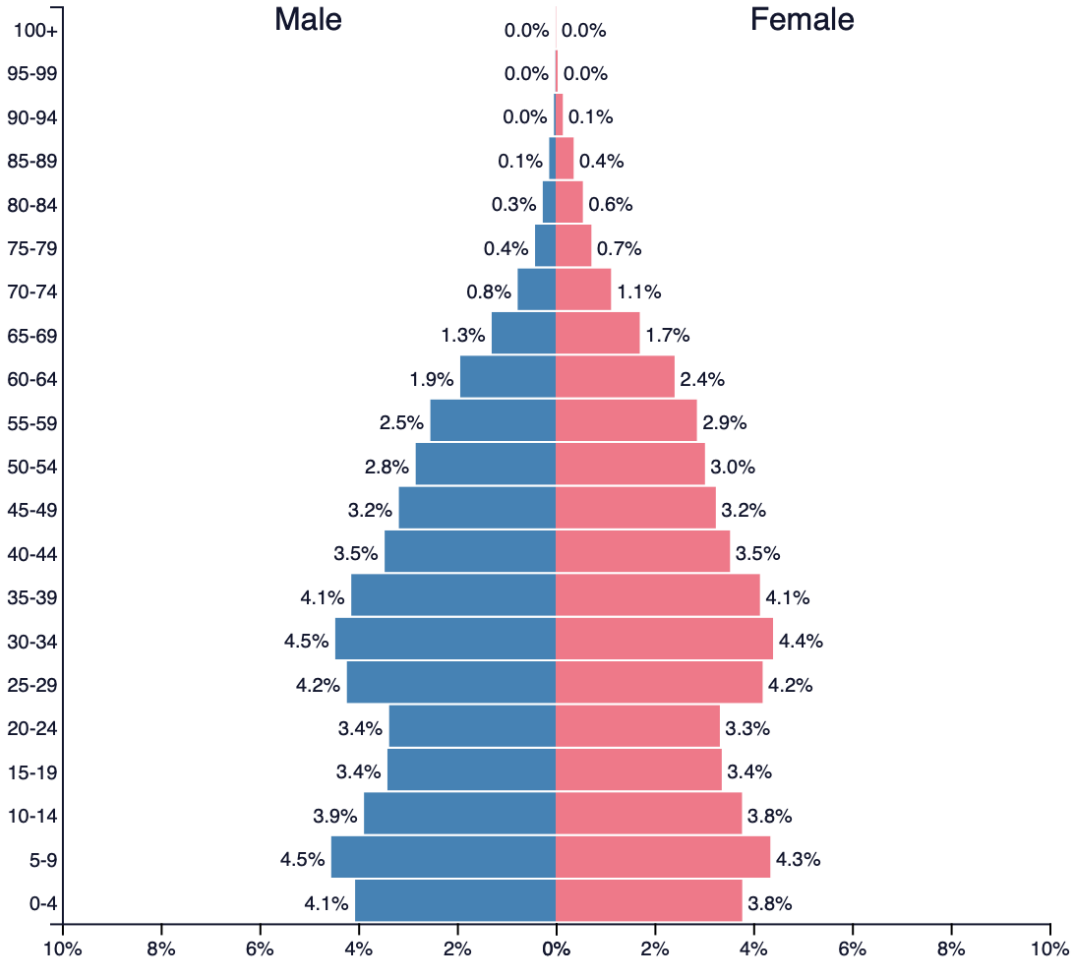
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Canada and Vietnam: Population Pyramid, 2020

The mathematical wizardy underpinning demographic analysis: every year, you get one year older



Canada – 2020
Population: 38,171,901

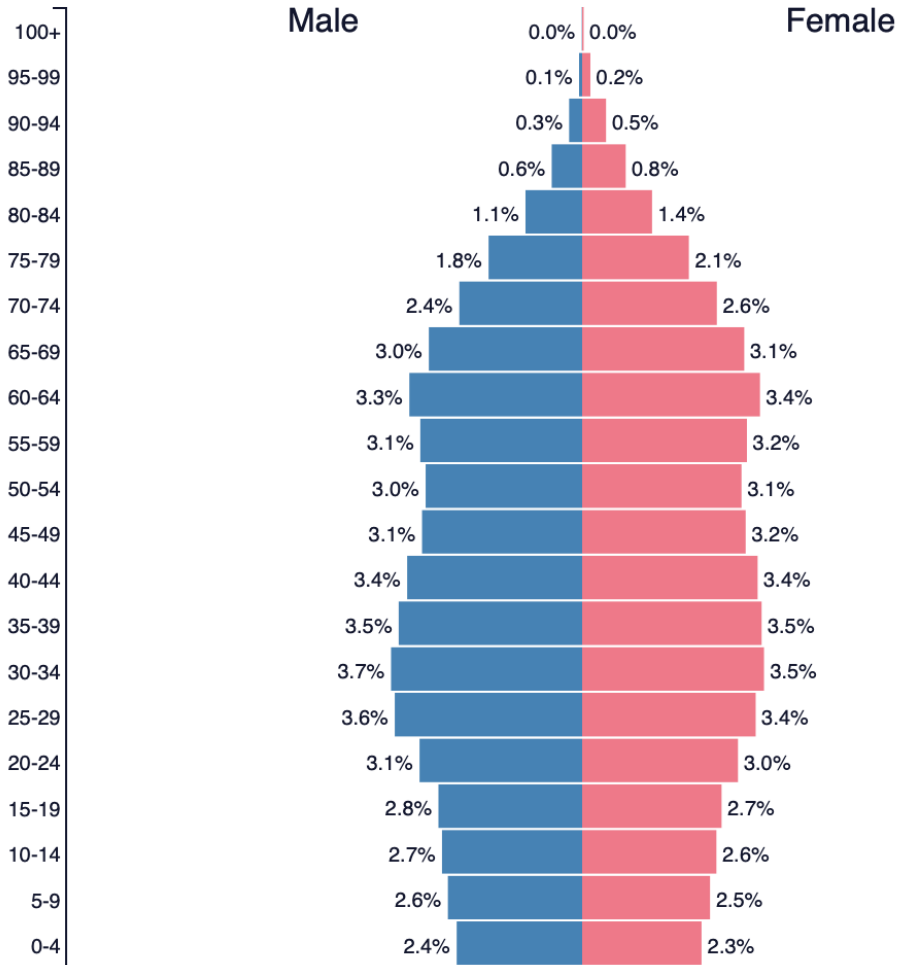


Vietnam – 2020
Population: 98,079,191

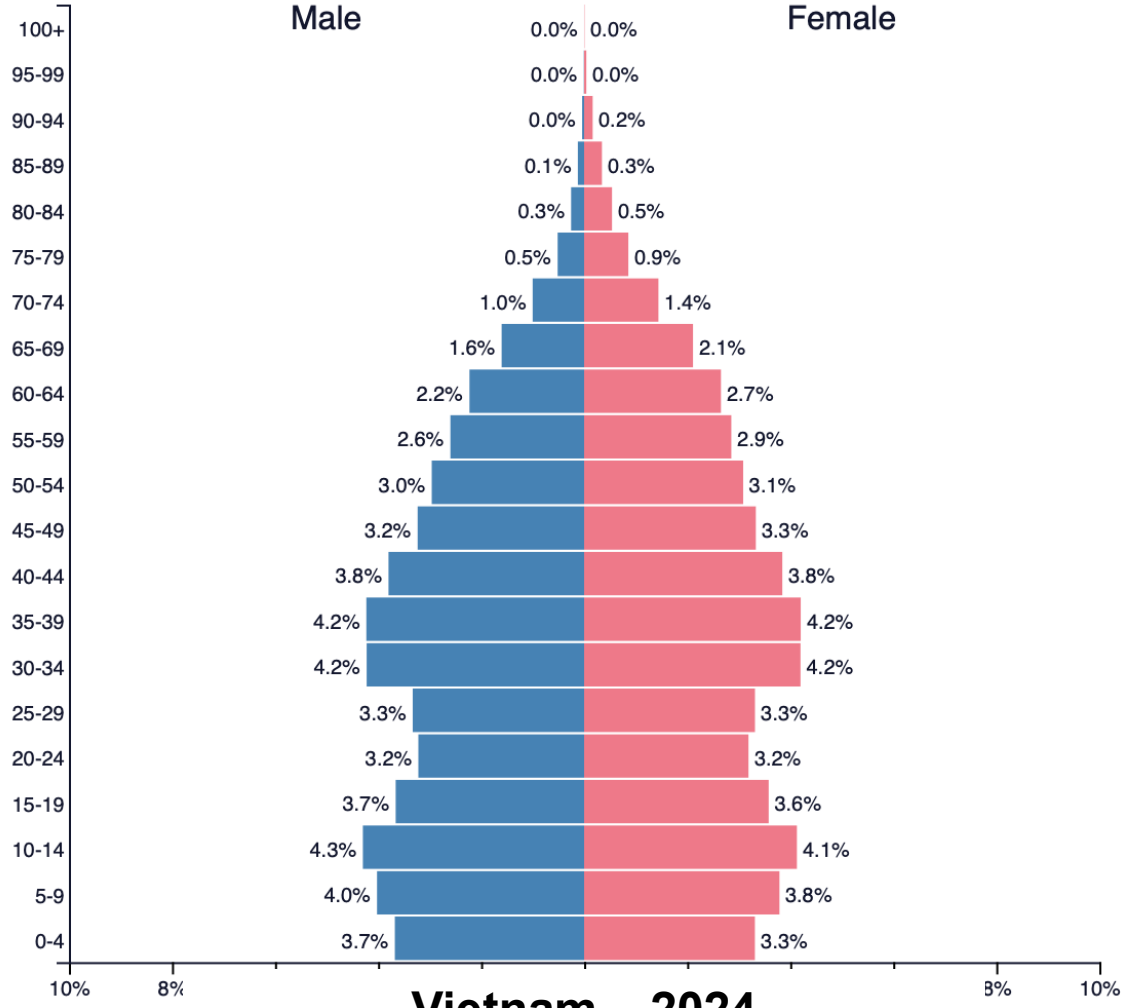
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Canada and Vietnam: Population Pyramid, 2024

Canada: from pyramid to kite. Vietnam: from pyramid to…temple?



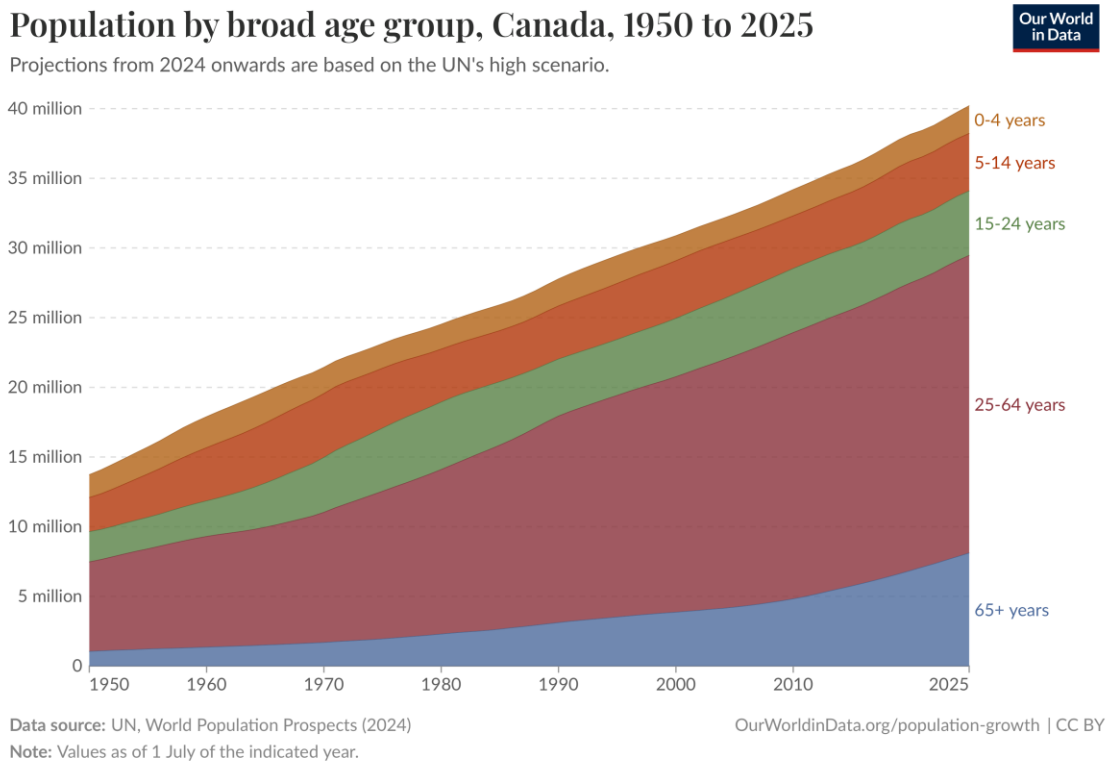
Canada – 2024
Population: 39,742,430



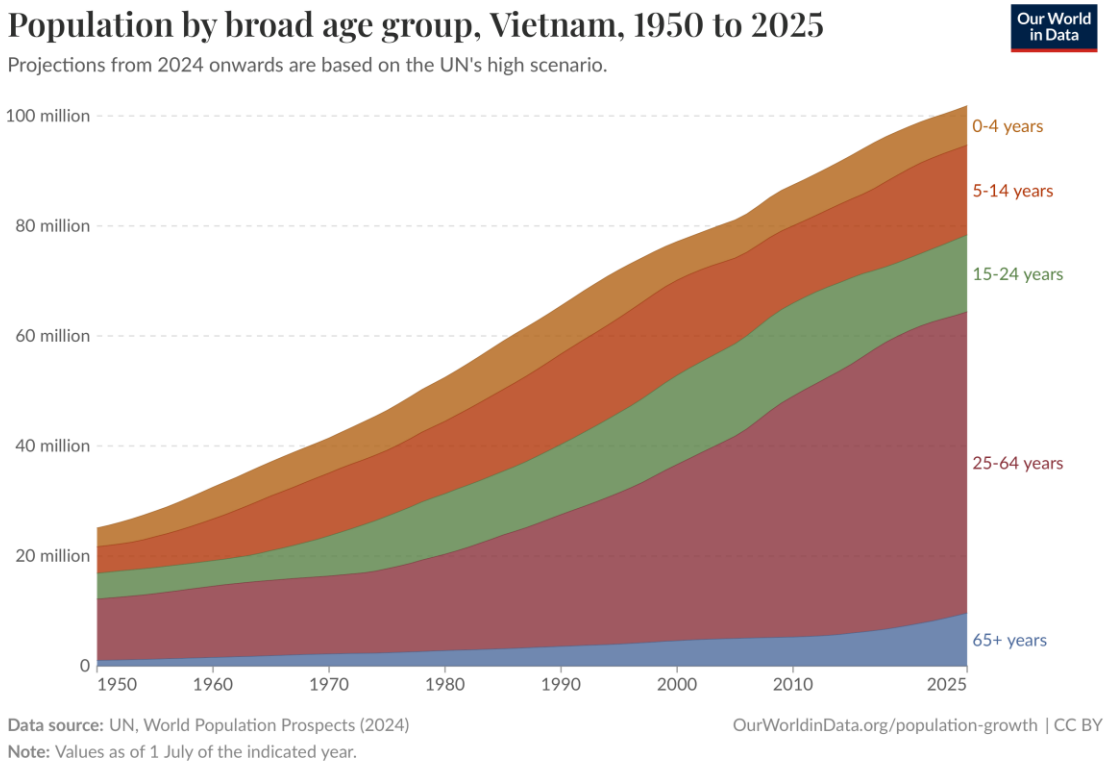
Vietnam – 2024
Population: 100,9879,686

Canada and Vietnam: Demographics

Add workers. Grow economy. Duh.



- 17 million people 25-64 in 2000
- 21 million people 25-64 in 2025
- (+24%)

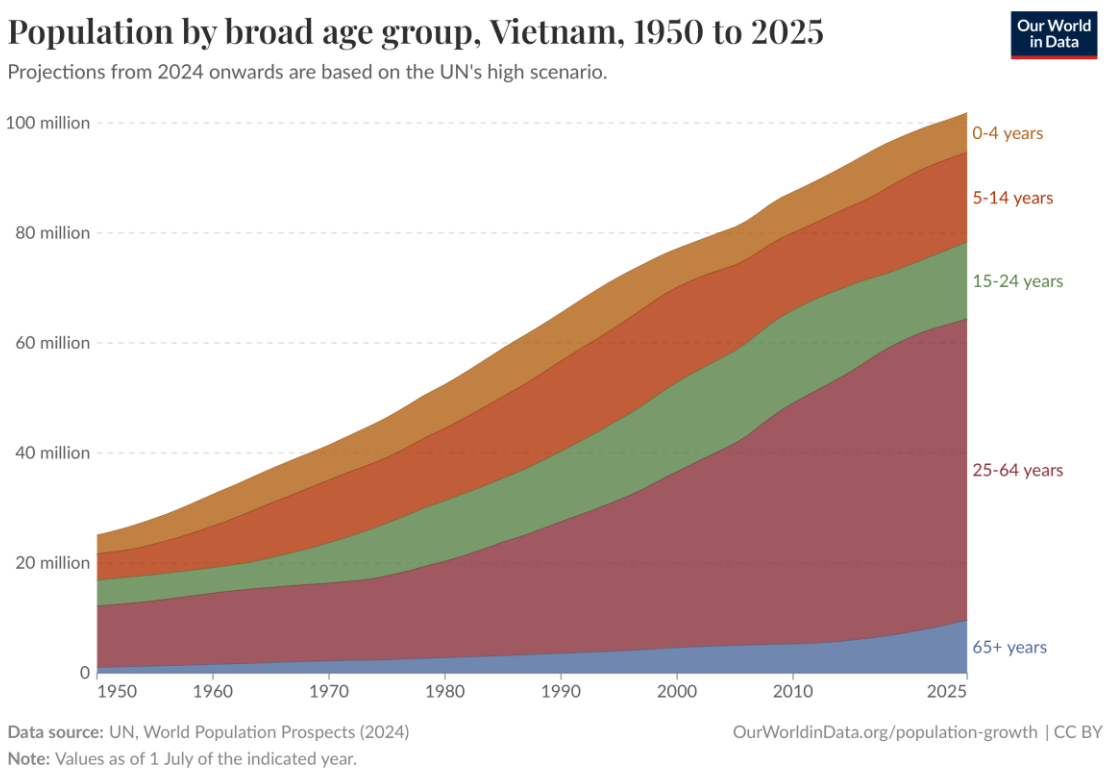


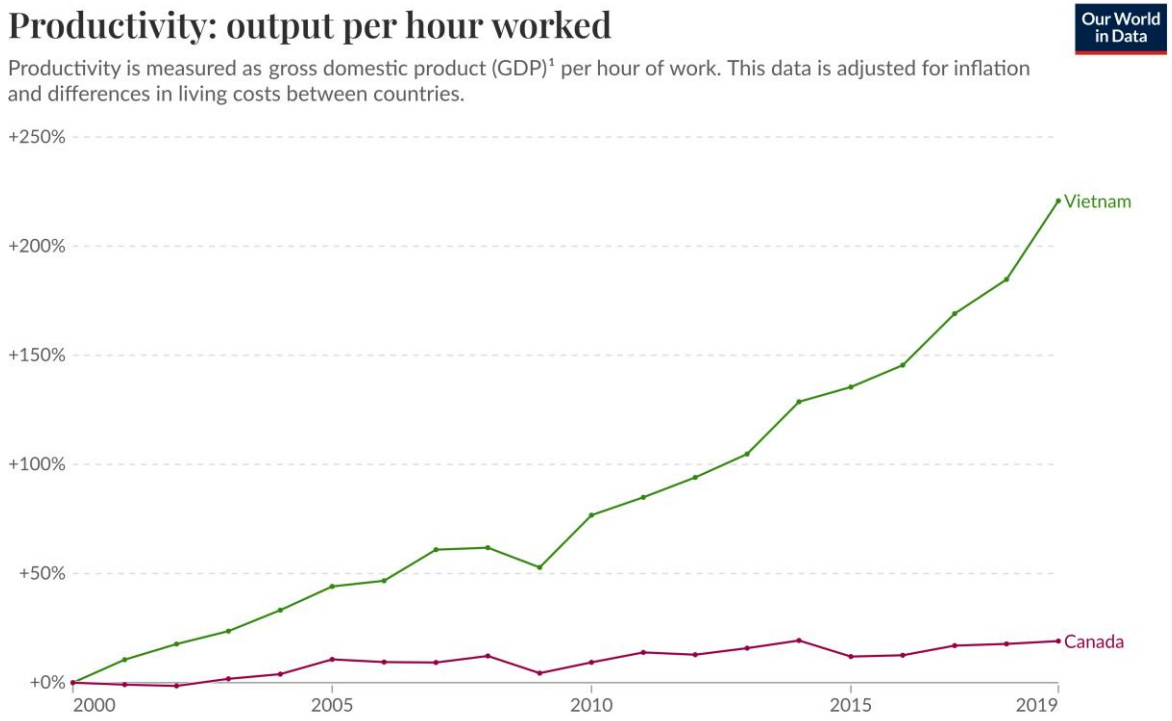
- 32 million people 25-64 in 2000
- 53 million people 25-64 in 2025
- (+65%)

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Summary: Our Canada-Vietnam growth story

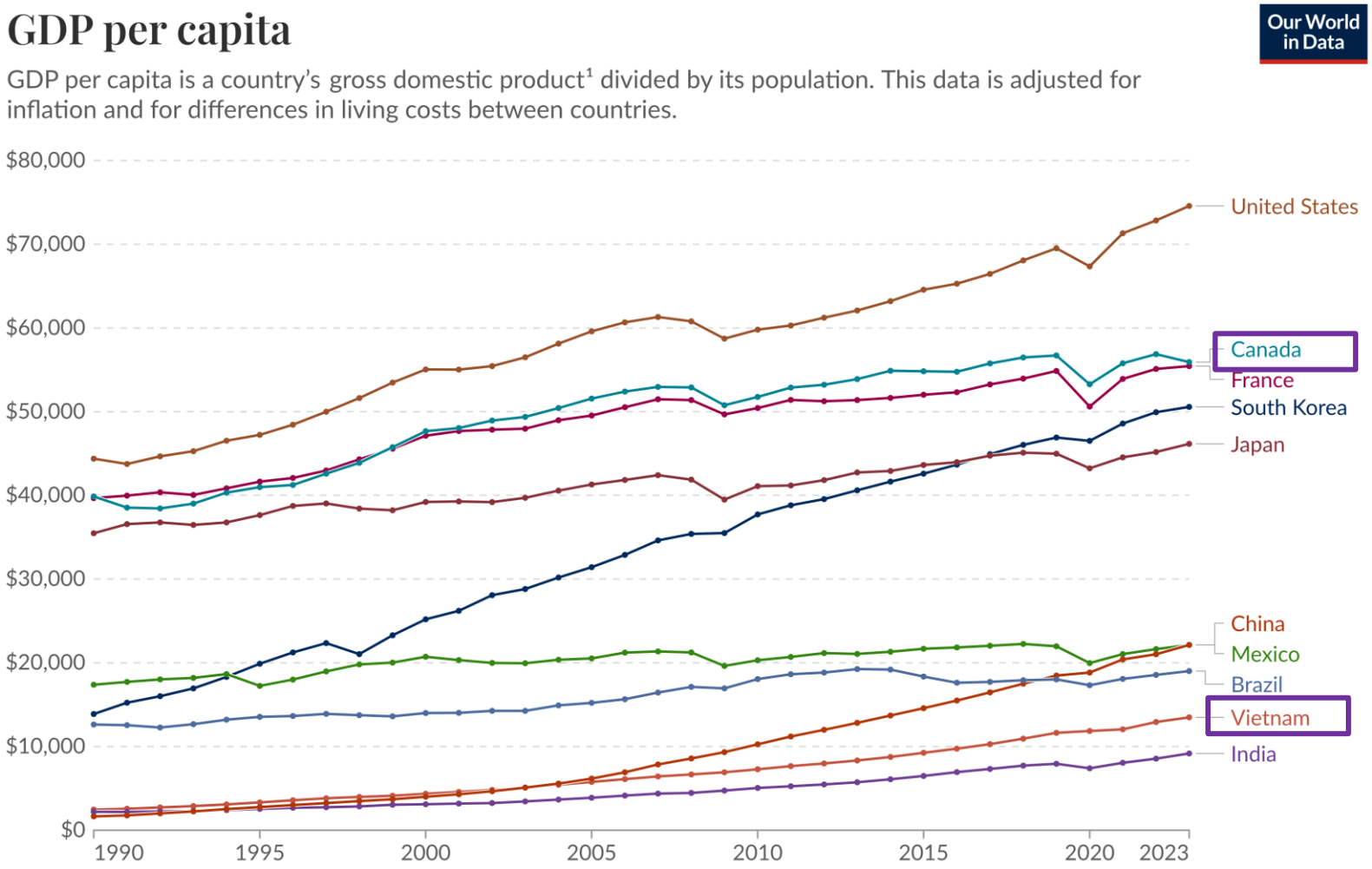
Vietnam: faster growth rate in working age population + higher growth rate in output per worker
= faster trend growth rate





Vietnam (et al): fast growth *and* still lots of runway

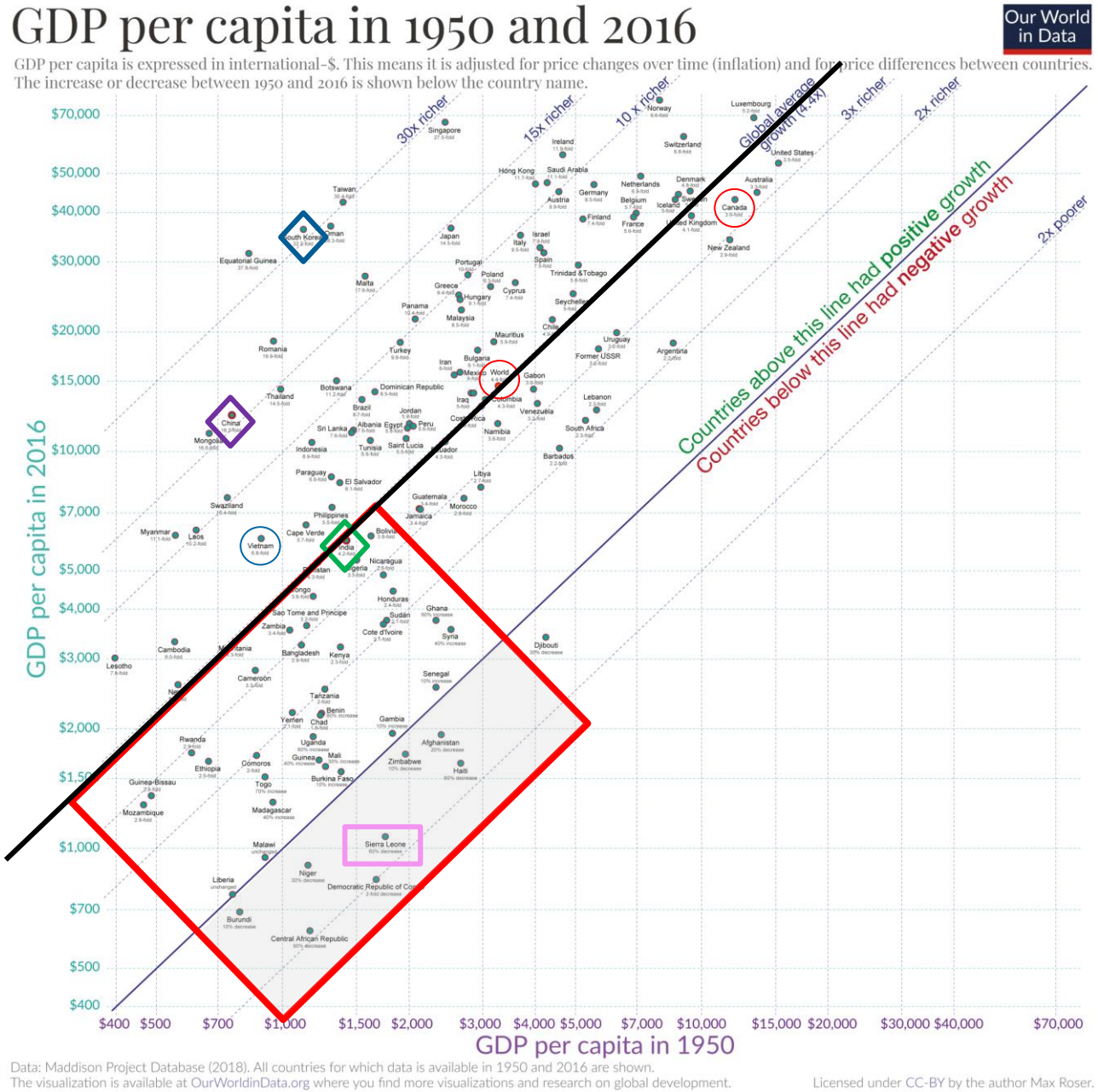
Vietnam experienced rapid growth over the last 35 years…but it is still a relatively poor country with lots of growth opportunities in front of it. Canada? We've picked the low hanging fruit. The result: Vietnam is very likely to continue to catch up to Canada, a good thing. This is known as the "convergence hypothesis".



The convergence hypothesis: do poorest grow fastest?

Overall, yes, but not as robustly as we should expect…or desire

This set of the poorest countries in 1950 grew at below world average rates. And all of the negative growth countries were very poor in 1950.

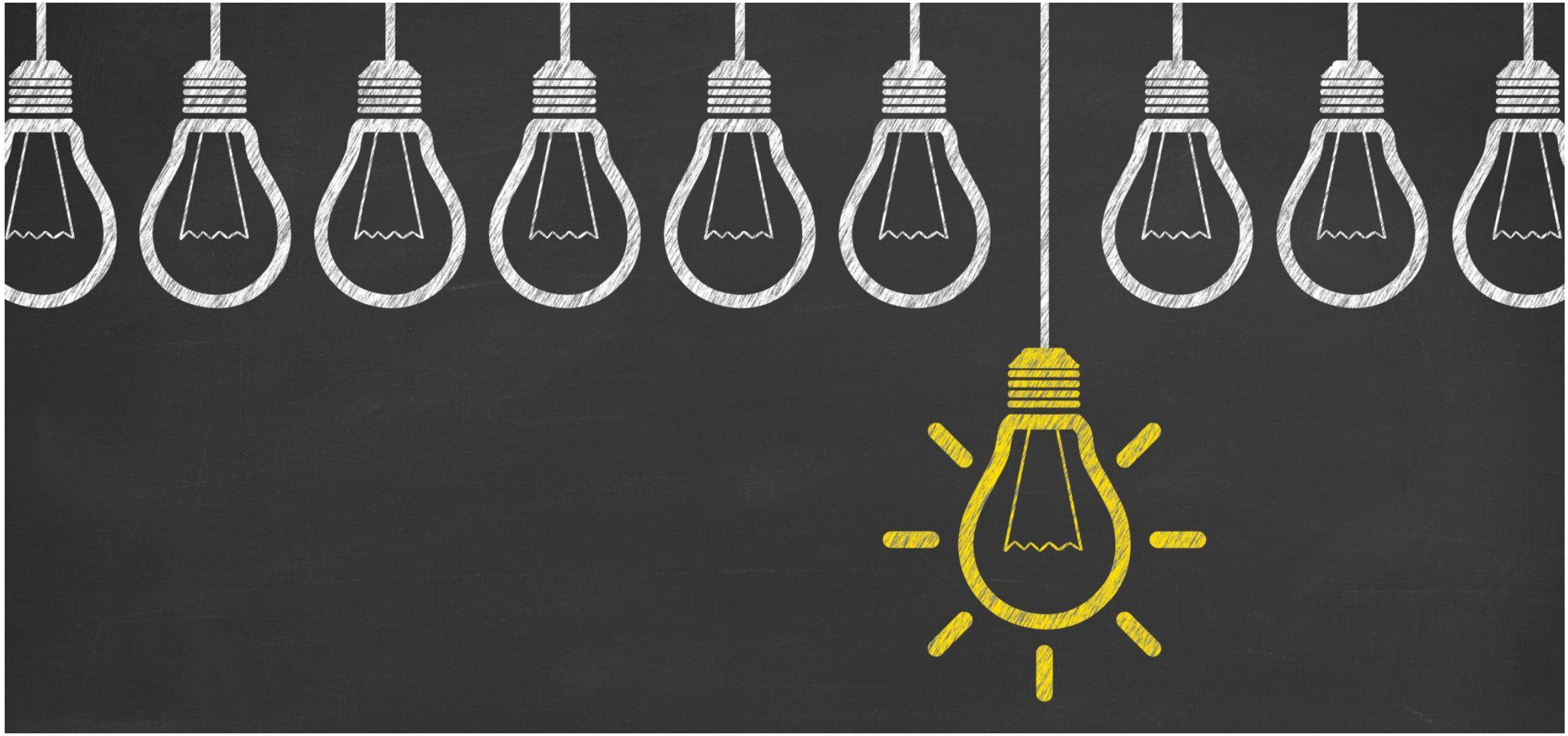


Vietnam: What could go wrong?

Might want to ask Canada about that…



GMM framework for trend growth



Motivation: A simple GDP decomposition

$$GDP = Y$$

Motivation: A simple GDP decomposition

$$Y = \left(\frac{Y}{Population}\right) * Population$$

Motivation: A simple GDP decomposition

$$Y = \left(\frac{Y}{Labour\ Hours}\right) * \left(\frac{Labour\ Hours}{Population}\right) * Population$$

Motivation: A simple GDP decomposition

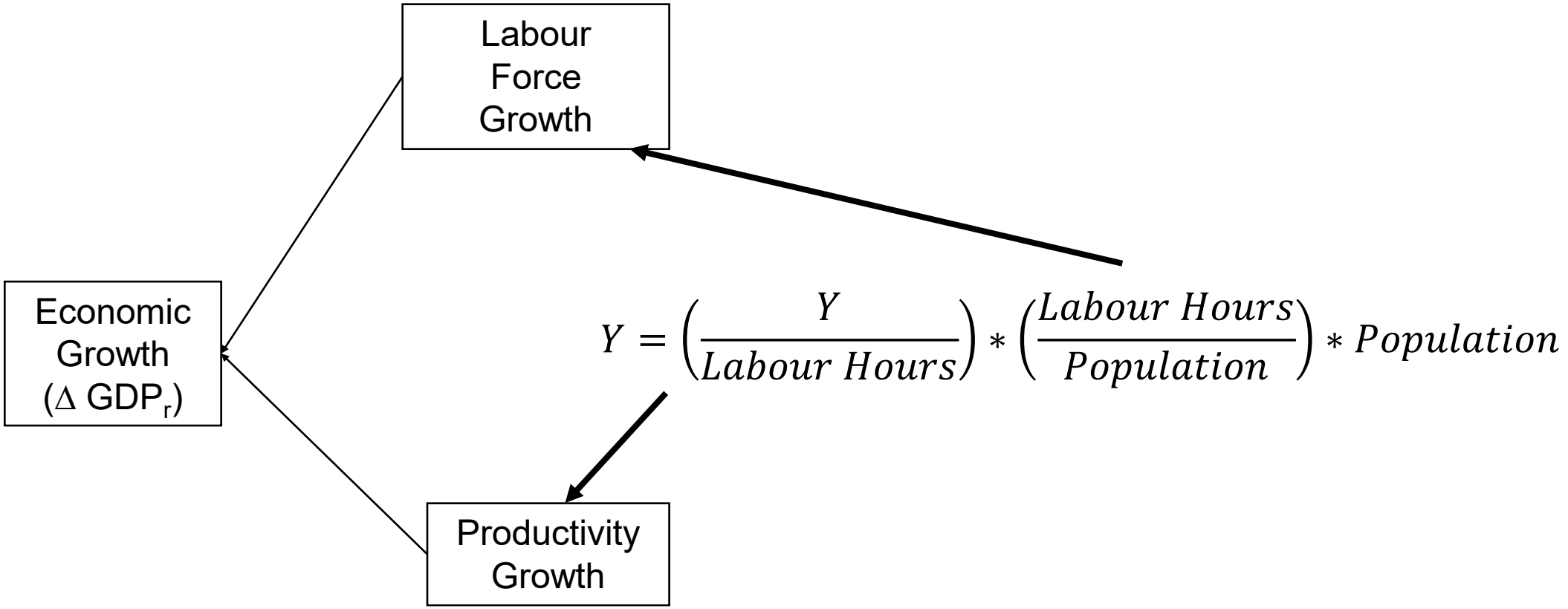
$$Y = \left(\frac{Y}{Labour\ Hours}\right) * \left(\frac{Labour\ Hours}{Population}\right) * Population$$

$Labour\ Force$

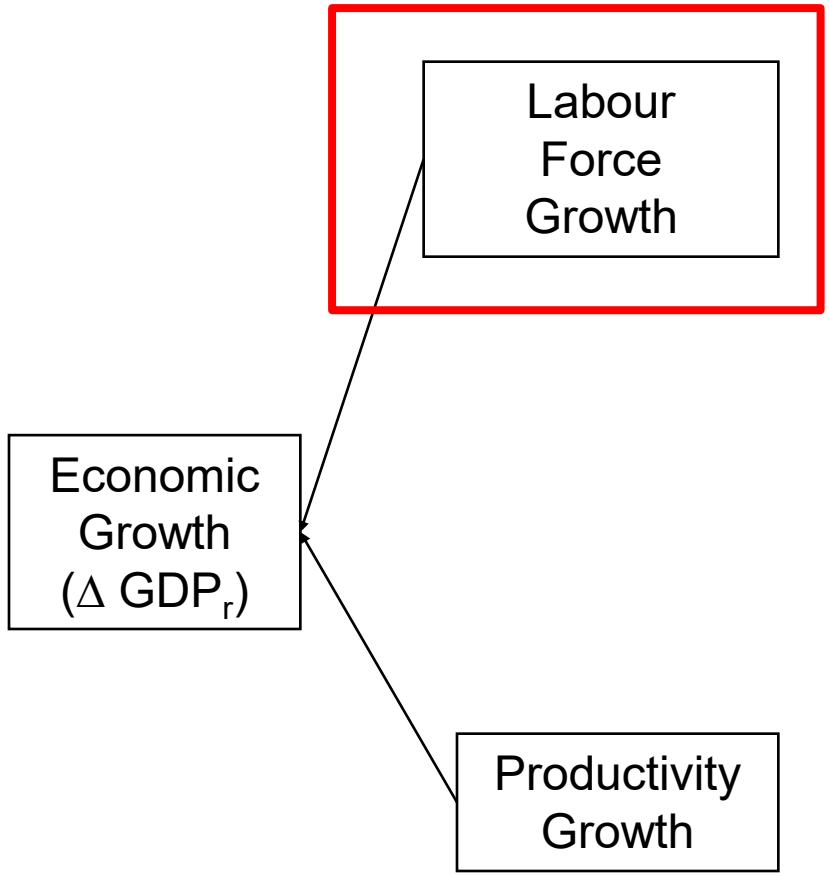
$Labour\ Productivity$

What does our case study of Vietnam tell us about where most of the growth comes from?

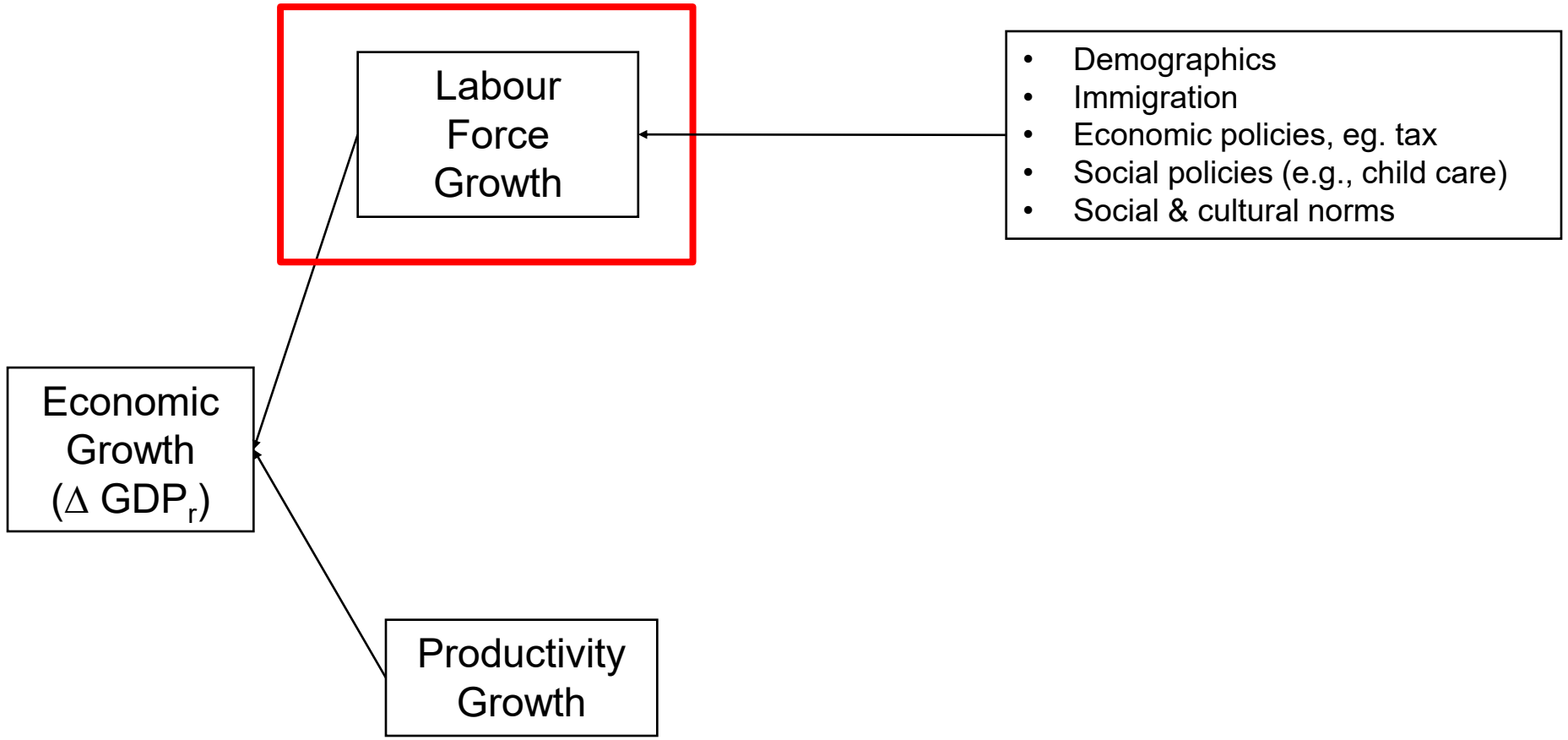
Trend Growth: Supply side determinants



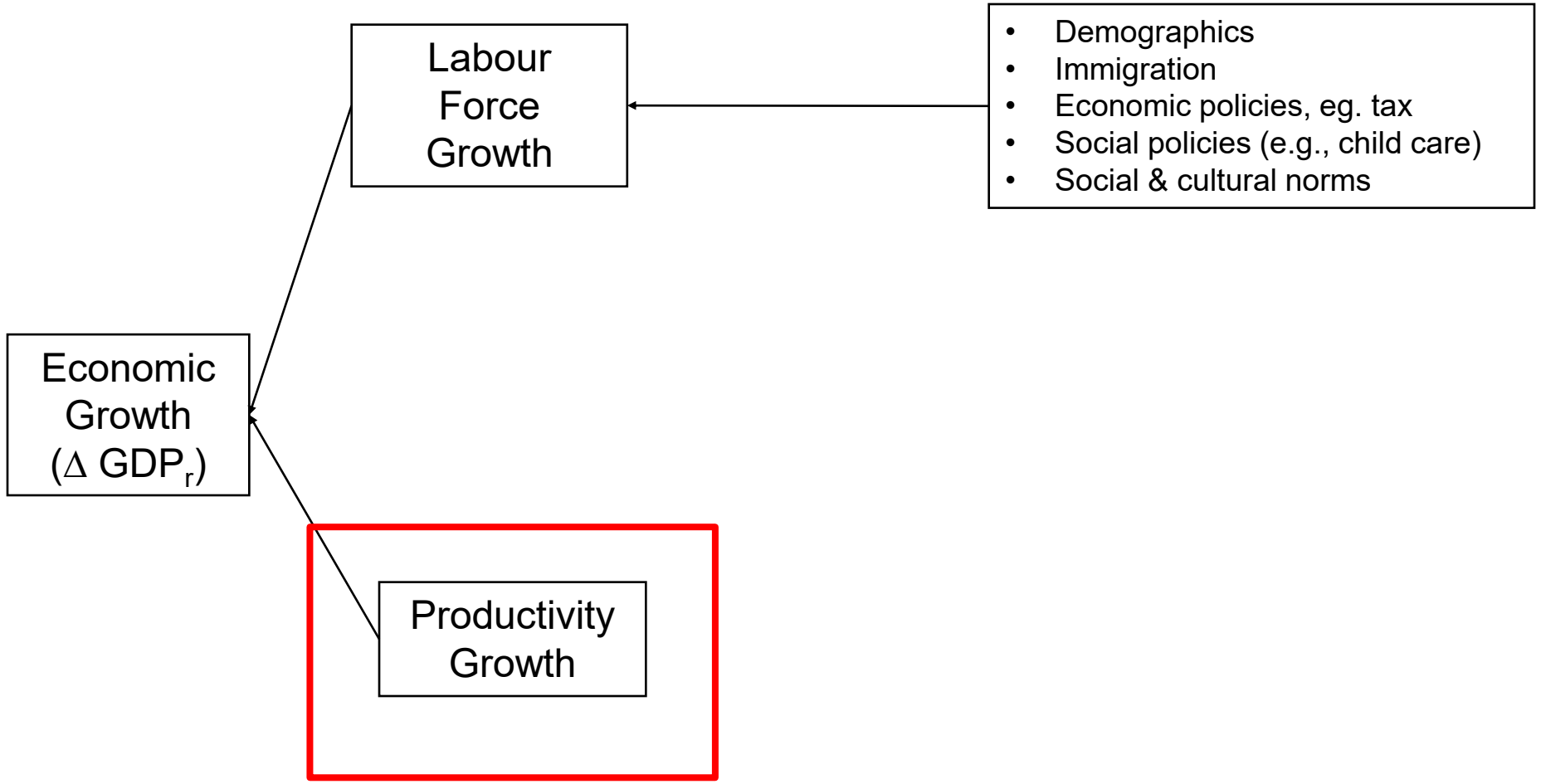
Trend Growth: Supply side determinants



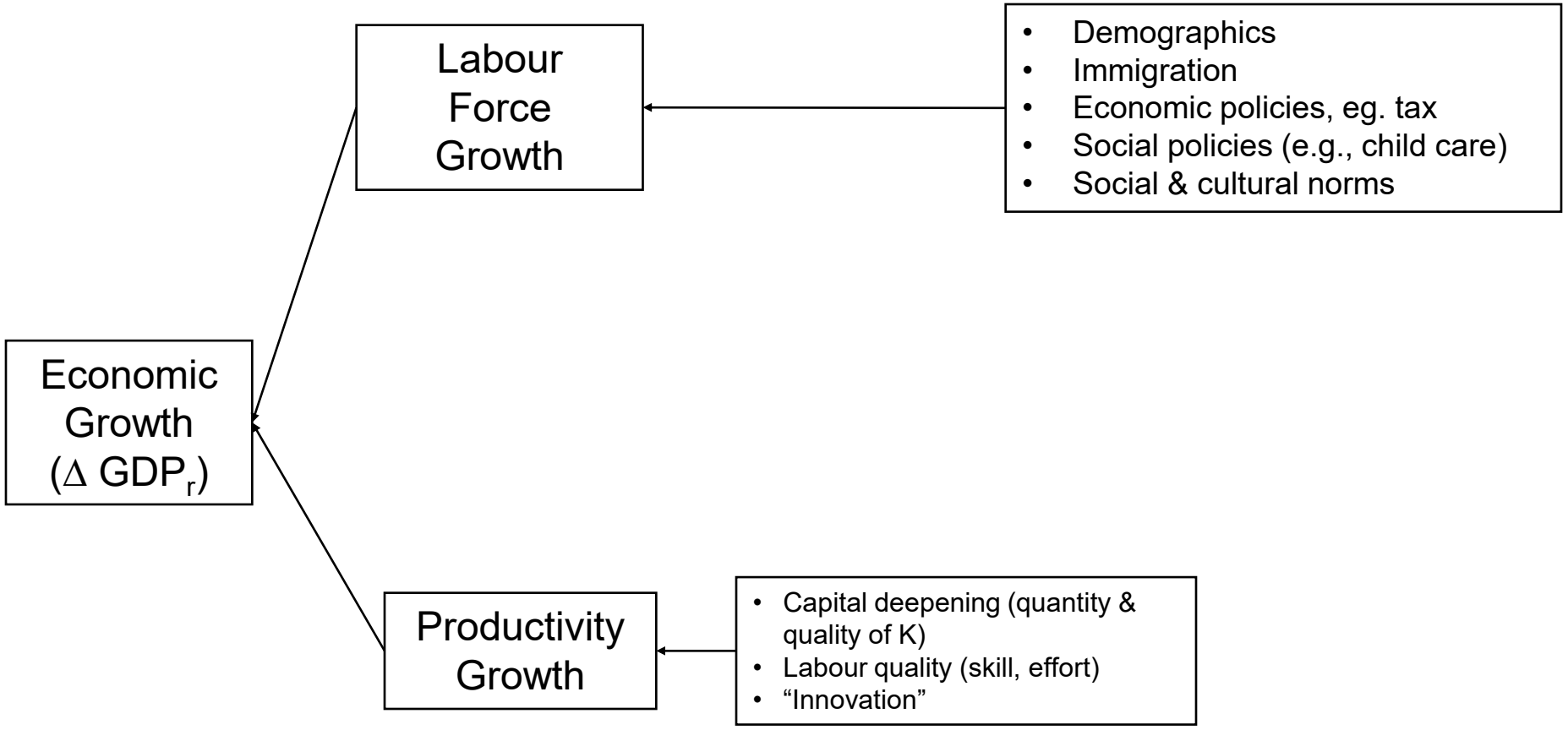
Trend Growth: Supply side determinants



Trend Growth: Supply side determinants

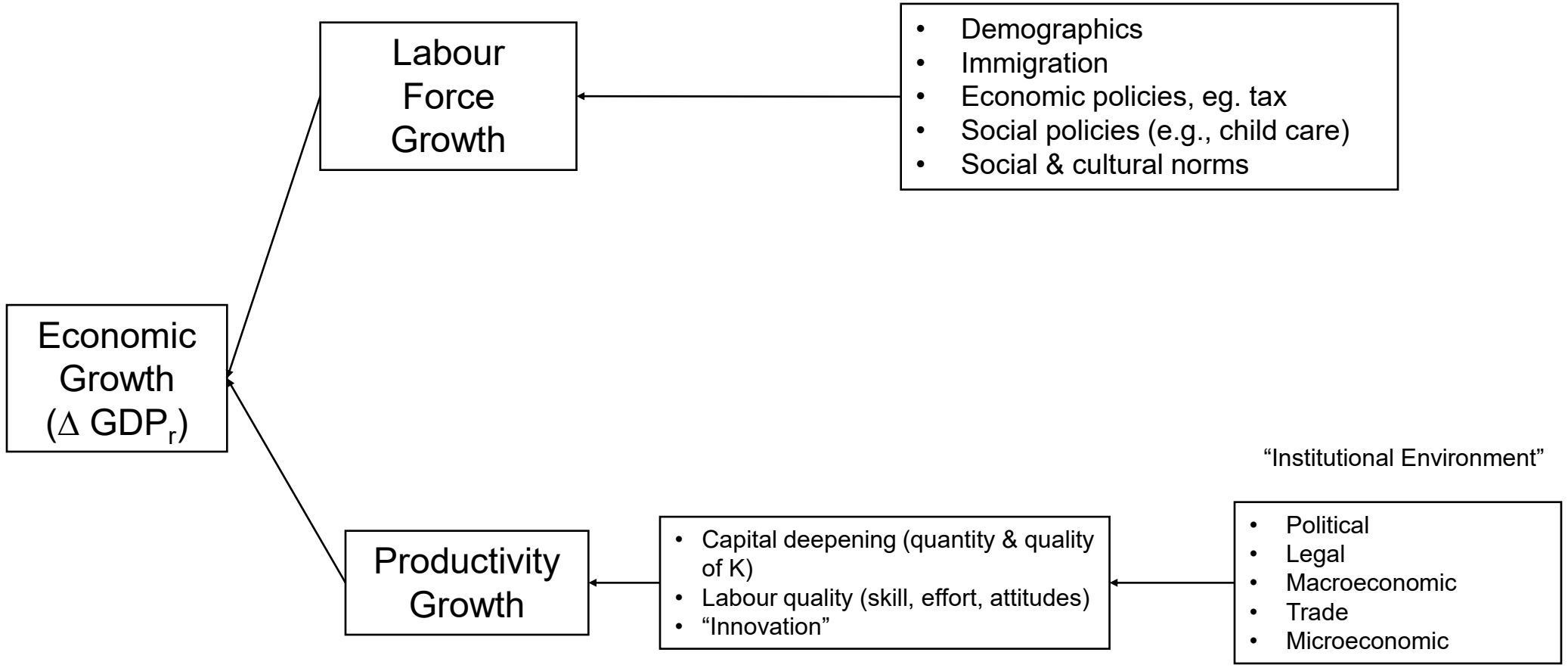


Trend Growth: Supply side determinants



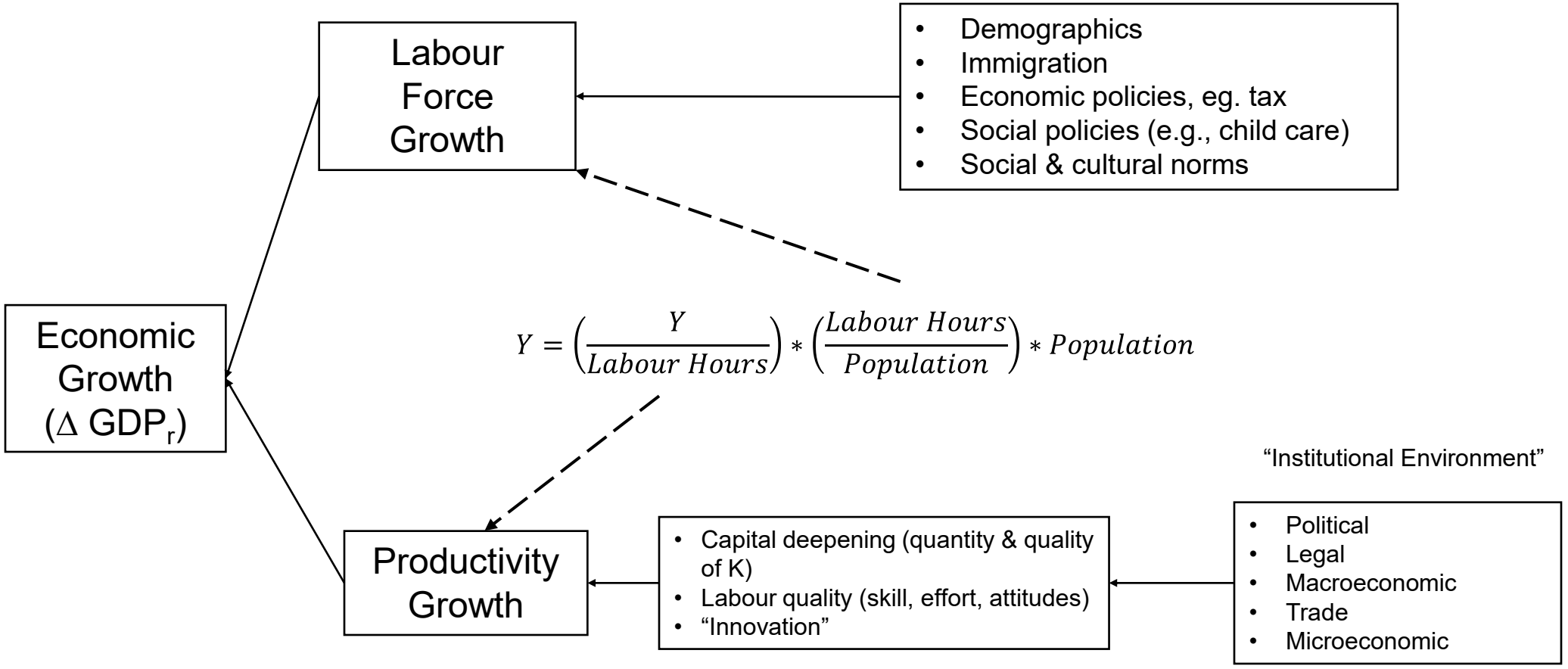
Trend Growth: Supply side determinants

Proximate and root causes of differences in trend growth rates across place and time



Trend Growth: Supply side determinants

Proximate and root causes of differences in trend growth rates across place and time



Back to the GMM meta chart

So, when we talk about “trend growth”, what we mean is: the growth in labour supply and labour productivity over time. The red line represents our capacity, or potential output. And it grows over time.

